NAVAL AIR TRAINING COMMAND

NAS CORPUS CHRISTI, TEXAS
Q-2A-6175

CNATRAINST 1542.175D
10 Jul 2025

CHIEF OF NAVAL AIR TRAINING

T-44C E-2D INTERMEDIATE MULTI-SERVICE PILOT TRAINING SYSTEM (MPTS) MASTER CURRICULUM GUIDE

2025

DEPARTMENT OF THE NAVY
CHIEF OF NAVAL AIR TRAINING
250 LEXINGTON BLVD SUITE 179
CORPUS CHRISTI TX 78419-5041

CNATRAINST 1542.175D CH-1
N713
11 Mar 2026

CNATRA INSTRUCTION 1542.175D CHANGE TRANSMITTAL 1

From: Chief of Naval Air Training

Subj: T-44C E-2D INTERMEDIATE MULTI-ENGINE MULTI-SERVICE PILOT TRAINING SYSTEM (MPTS) FLIGHT TRAINING CURRICULUM

Encl: (1) Revised pages 3.
(2) Revised page v.
(3) Revised page ix.
(4) Revised pages IV-13 – IV-14.
(5) Revised page V-20.

1. Purpose

a. Removed the FAM4206 Special Syllabus Requirement.

b. Modified FAM42 block by reducing FAM4206 hours from 2.0 hrs to 1.5 hrs. Total block hours are now 10.0.

c. Added a syllabus note to I43 block.

2. Action. This curriculum is effective on receipt. No changes will be made without written authorization by the Chief of Naval Air Training (CNATRA).

a. Remove pages 3, v, ix, IV-13, IV-14, and V-20.

b. Insert enclosure (1).

c. Enter the date of Change 1 on the new Summary of Changes page.

3. Records Management. Records created as a result of this instruction, regardless of media and format, must be managed per Secretary of the Navy Manual 5210.1 of September 2019.

4. Review and Effective Date. Per this instruction, OPNAVINST 5215.17A, CNATRA N7 will review this instruction annually around the anniversary of its effective date to ensure applicability, currency, and consistency with Federal, DoD, SECNAV, and Navy policy and statutory authority using OPNAV 5215/40 Review of Instruction. This instruction will be in effect for 10 years, unless revised or cancelled in the interim, and will be reissued by the 10-year anniversary date if it is still required, unless it meets one of the exceptions in OPNAVINST 5215.17A paragraph 9. Otherwise, if the instruction is no longer required, it will be processed

for cancellation as soon as the need for cancellation is known following the guidance in OPNAV Manual 5215.1 of May 2016.

5. Forms. The CNATRA forms required by this instruction are automated in the Training Learning Management System (T/LMS) computer program. Additional copies of CNATRA forms are available on the CNATRA Web site: https://www.cnatra.navy.mil/pubs-forms.asp.

P. N. FLORES
Chief of Staff

Releasability and distribution:
This change transmittal is cleared for public release and is available electronically only via Chief of Naval Air Training Issuances website, https://cnatra.navy.mil/pubs-secure.asp.

DEPARTMENT OF THE NAVY
CHIEF OF NAVAL AIR TRAINING
250 LEXINGTON BLVD SUITE 179
CORPUS CHRISTI TX 78419-5041

CNATRAINST 1542.175D
N716
10 Jul 2025

CNATRA INSTRUCTION 1542.175D

From: Chief of Naval Air Training

Subj: T-44C E-2D INTERMEDIATE MULTI-SERVICE PILOT TRAINING SYSTEM FLIGHT TRAINING CURRICULUM

1. Purpose. To publish the curriculum for training Naval Aviators in the E-2D Intermediate Multi-Service Pilot Training System phase of flight training.

2. Cancellation. CNATRAINST 1542.175C will be cancelled when the last student enrolled completes the curriculum or is transitioned to the CNATRAINST 1542.175D.

3. Action. This curriculum is effective on receipt. No changes will be made without written authorization by the Chief of Naval Air Training (CNATRA).

4. Records Management. Records created as a result of this instruction, regardless of media and format, must be managed per Secretary of the Navy Manual 5210.1 of September 2019.

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6. Forms. The CNATRA forms utilized with this instruction are automated in the Training Learning Management System (T/LMS) computer program. Additional copies of CNATRA forms are available on the CNATRA Web site: https://www.cnatra.navy.mil/pubs-forms.asp.

P. N. FLORES
Chief of Staff

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LIST OF EFFECTIVE PAGES

Total number of pages is 132 consisting of the following:

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iii/(iv blank)	
v/(vi blank)	1
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xix/(xx blank)	
I-1 – I-14	
II-1 – II-12	
III-1/(III-2 blank)	
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1		Incorporated	3, v, ix, IV-13, IV-14, V-20

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COURSE DATA

1. Course Title. T-44C E-2D Intermediate Multi-Service Pilot Training System (MPTS) Flight Training Curriculum.

2. Course ID Number (CIN). Q-2A-6175.

3. Location. Naval Air Station, Corpus Christi, Texas 78419.

4. Course Status. Active.

5. Course Mission. The mission of E-2D Intermediate MPTS is to develop multi-engine flight skills while emphasizing instrument flying and crew coordination. At the successful completion of this phase of aviation training, the student will enter E-2D Advanced MPTS training.

6. Prerequisite Training. Successful completion of:

a. T-6B Joint Primary Pilot Training (JPPT), (TW4: Q-2A-3166 or TW5: Q-2A-4166) or T-6B Primary Pilot Training, Naval Aviation Training Next (Avenger) Syllabus (TW4: Q-2A-8195 or TW5: Q-2A-9195) or,

b. T-6B Joint Primary Pilot Training (JPPT), CNATRAINST 1542.166D.

7. Security Clearance Required. None.

8. Follow-on Training. As required by Chief of Naval Air Training (CNATRA) for each specific mission assignment.

9. Course Length. Overall time-to-train is calculated as per the CNATRAINST 1550.6G. Training Days account directly or provide margin for factors including weather, personnel and equipment availability, briefing and preparation time, and historical delays. Production Weeks further account for weekends, holidays, safety stand-downs, and other expected
nonworking days throughout the year.

	Training Days	Production Weeks
T-44C E-2D Intermediate MPTS	85.2	19.0

10. Class Capacity. Variable.

11. Instructor Requirements. As established by Chief of Naval Operations (CNO) planning factors.

12. Course Curriculum Model Manager. Commander, Training Air Wing FOUR (COMTRAWING FOUR).

13. Quota Management Authority. CNATRA.

14. Quota Control. CNO.

15. Course Training Subjects

a. Ground Training

GROUND TRAINING		
Subject	**Symbol**	**Hours**
Indoctrination	G01	4.1
Systems	G02/G03	31.5
Meteorology	G04	11.0
Aerodynamics	G05	24.5
Flight Procedures	G06	9.25
Crew Resource Management	G07	3.0
Instrument Flight Rules – Phase I	G08	25.5
Instrument Flight Rules – Phase II	G09	21.5
Flight Rules and Regulations	G10	13.5
Total		**143.85**

b. Flight Support

FLIGHT SUPPORT		
Subject	**Symbol**	**Hours**
Checklist Orientation	FAM01	2.0
Familiarization Brief	FAM02	5.0
Instrument Brief/Flight Management System	I01	7.0
Maritime Formation Procedures	F01	1.0
Total		**15.0**

c. Flight Training. Below are the programmed times for each phase, stage, and media:

FLIGHT TRAINING									
Block	Flight/Events	OFT		UTD		T-44C			
						Dual		Solo	
		Flts	Hrs	Flts	Hrs	Flts	Hrs	Flts	Hrs
FAM21	Familiarization Procedures	1	1.5	4	6.0				
FAM31	Familiarization	1	1.5						
FAM41	Familiarization					4	8.0		
FAM32	Emergency Familiarization	1	1.5						
FAM42	Emergency Familiarization					6	10.0		
FAM33	Familiarization EP/CRM	1	1.5						
FAM43	Familiarization Check Flight					1	1.5		
FAM34	Night Familiarization	1	1.5						
FAM45	Night Familiarization					2	3.0		
I21	Basic Instruments			3	4.5				
I31	Instrument Approaches	4	6.0						
I41	Instrument Approaches					4	8.0		
I32	Emergency Instrument Approaches	4	6.0						
I42	Emergency Instrument Approaches					4	8.0		
I33	Mid-Stage Instrument Check	1	1.5						
I43	Cross Country					4	16.0		
I34	Advanced Emergency Instrument Approaches	2	3.0						
I35	Instrument EP/CRM	2	3.0						
I44	Advanced Emergency Instrument Approaches					3	6.0		
I45	NATOPS Instrument Check Flight					1	2.0		
F41	Maritime Formation Fundamentals					1	2.0		
Total		**18**	**27.0**	**7**	**10.5**	**30**	**64.5**	**0**	**0**

16. Training Preparation Time. In addition to the hours formally planned for classes, simulators, and flights, significant additional time to prepare and study should be expected outside of scheduled training hours. This range will vary depending on the complexity of the material and individual student needs, and may be up to several hours per event. For simulator and flight events, specific brief and taxi times will be programmed into T-SHARP and accounted for on the flight schedule, per the following table:

ADDITIONAL FORMAL TRAINING TIME PER EVENT			
Training Area	**Brief/ Preflight/ Taxi**	**Taxi/ Debrief**	**Total**
Flight	2.25	1.0	3.25
Simulator/CPT Dual Event	1.0	1.0	2.0
Simulator/CPT Single Event	0.5	0.5	1.0

17. Physical Requirements. As specified in the Manual of the Medical Department (NAVMED P-117) and all applicable anthropometric standards.

18. Obligated Service. Refer to MILPERSMAN for Naval personnel.

19. Primary Instructional Methods. Lecture, computer aided instruction (CAI), self- and group-paced study, and in-flight instruction.

20. Preceding Curriculum Data. This curriculum replaces the E-2D curriculum in CNATRAINST 1542.175C.

21. Student Performance Measurement and Application of Standards. The standards outlined in Chapter IX, Course Training Standards, are used to evaluate Student Naval Aviator (SNA) performance of individual items and maneuvers. Final judgment regarding the satisfactory performance of any flight maneuver rests with the Instructor Pilot who must assess the environmental and systems factors affecting the conditions under which the performance is measured and the SNA's experience within the stage.

GLOSSARY

1. Advancing X. Completed event within the normal syllabus flow. Excludes events with last characters in the range 84-89.

2. Aviation Training Form (ATF). A grade sheet documenting SNA performance for all categories of training regardless of media, phase, or stage.

3. Aviation Training Jacket (ATJ). The ATJ is the SNA's training record. It contains ATFs, calendar card, grade reports, and all other associated training information. It is filed in student control and follows the SNA through all phases of training.

4. Aviation Training Summary (ATS). A tabular sheet listing the MIF and maneuver grades within a training stage.

5. Block of Training. A sequential series of lessons within a training stage sharing an identical MIF. The third character in the lesson designator identifies a block.

6. Check Flight (SXX90). A flight check in any stage of training.

7. Course of Training. The entire program of preflight, flight, simulation, academics, and officer development conducted in all media during the programmed training days.

8. Course Training Standard (CTS). A description of required behaviors and standards of performance for a specific maneuver. These standards are in Chapter IX.

9. Courseware. The technical data, flight training instructions, audio, video, film, CAI, instructor guides, student study guides, and other training material developed to support and implement the syllabus of instruction.

10. Critical Item. Any maneuver coded with a plus sign (+). This symbol indicates the maneuver is required and must be accomplished to the specified standard in that block of training.

11. Commanding Officer Progress Check (CO-PC) (SXX89). A special check normally given by the CO or XO. The CO may designate, in writing, CO-PC duty to a qualified O-4 or above. This is only done if the CO or XO is unqualified or unavailable to instruct in the required stage. A satisfactory CO-PC returns the student to normal syllabus flow. An UNSAT CO-PC results in a TRB.

12. Deliverables. A CNATRA 1542/1827 (Rev. 4-04) Training Review Board Summary Form, generated by the TRB, which summarizes a specific SNA's progress in a given syllabus and provides detailed information on the application of MPTS training for that SNA. Deliverables indicate whether the quality and continuity of training provided was per CNATRAINST 1542.175D and indicate the degree of influence by "human factors" on the SNA's performance.

13. Demonstrate. Instructor performs the maneuver with precision and accompanying description. SNA is responsible for knowledge of the procedures prior to event brief and observes the maneuver.

14. Emergency Procedures (EP). Any degradation of aircraft systems or flight conditions requiring pilot action or intervention.

15. End of Block (EOB). Last event in block. In order to progress past EOB, the SNA must meet or exceed MIF on all critical items, and all optional items attempted, by the end of the block. Flight shall consist of a cross-section of critical items; however, all critical items do not have to be accomplished on the last flight in block as long as MIF had been previously met.

16. Extra Training (SXX87). Additional SNA training flights ordered by the Commanding Officer, or higher, in order to compensate for documented instructional deficiencies.

17. Familiarization. The stage of training that includes both day and night familiarization.

18. Flight Training Instruction (FTI). A CNATRA-approved manual describing flight procedures and techniques for each training stage.

19. Hours per X (H/X). The average length for each event in a block, rounded to the nearest tenth of an hour.

20. Initial Progress Check (IPC) (SXX88). A special check given by the most experienced instructors that have a complete understanding of NATRACOM and PC processes and understand the gravity of their responsibility in helping maintain the standards of Naval Aviation. These instructors should carry the CO's full confidence and shall be designated in writing by the CO. A satisfactory IPC returns the student to normal syllabus flow. An IPC can count as an Advancing "X" at the CO's discretion. An UNSAT IPC results in a CO-PC. IPC events shall be coded as XX88, (e.g., OPS3188).

21. Introduce. Instructor coaches the SNA through the maneuver as necessary and/or may demonstrate the maneuver again. The SNA is responsible for knowledge of the procedures prior to the event brief and for performing the maneuver with coaching.

22. Lesson Designator. All syllabus events have a five-character lesson designator in the following format:

Char	Meaning	Remarks		
1st	Stage	G – Ground FAM – Familiarization	I – Instrument F – Formation	
2nd	Media	0 or 1 – Ground Training/Flight Support	2 – CPT 3 – Simulator	4 – Aircraft
3rd	Block	Sequential, indicating block within stage.		
4th & 5th	Event/Check Identifier	Sequential, indicating event within block, or other event types as shown below: 84 – Adaptation 85 – Practice Sim 86 – Warmup 87 – Extra Training	88 – IPC 89 – CO-PC 90 – Check Flight/Exam	

23. Maneuver Item File (MIF). A listing of required maneuvers and associated proficiency levels for each block of training.

24. Master Syllabus. Chapters I-VII list all training syllabus activities, prerequisites, and training flow for MPTS.

25. Naval Air Training Class Advisor Program Officer. An Instructor Pilot assigned to provide counseling and guidance to a specific student pilot or pilots throughout the applicable syllabus.

26. Off-Wing Flight. A Familiarization flight not flown with the SNA's on-wing.

27. On-Wing. The SNA's assigned instructor in the Familiarization stage per CNATRAINST 1500.4L.

28. Phase of Training. A major division in the course of training. MPTS consists of three phases: Primary, Intermediate, and Advanced.

29. Pink ATF. A standard ATF that is printed on pink paper. The pink ATF is used to denote an UNSAT.

30. Practice. Instructor observes SNA with minimal coaching; may also demonstrate the maneuver if necessary. The SNA must perform maneuver with minimal coaching.

31. Progress Check Pilot. An Instructor Pilot authorized to administer Initial or Commanding Officer Progress Checks.

32. Ready Room UNSAT (RRU). An UNSAT grade given for inadequate knowledge of flight procedures, systems, discuss items, emergency procedures, or deficient preflight planning.

33. Special Syllabus Requirements (SSR). One-time, ungraded demonstration items. While SSRs are recommended for certain events in block, they may be accomplished at any time during the block.

34. Stage of Training. All training of a particular type (Ground, Familiarization, Instruments, Navigation, Formation, Tactical) within a phase. The first letter in the lesson designator identifies the stage of each lesson (Example: F4101 is in the Formation Stage).

35. Training Media. MPTS media include aircraft, simulator, CPTs, ground training, and CAI. The second character in the lesson identifier designates the training media.

36. Training Review Board (TRB). A fact-finding board generated by a failed CO-PC that considers the circumstances relevant to the SNA's training, such as quality and continuity of training, outside influences, and extenuating circumstances. The TRB does not make attrition or retention recommendations.

37. Warmup Event (SXX86). Additional events given to allow an SNA to regain a level of proficiency previously demonstrated which has diminished due to an extended break in training.

ABBREVIATIONS

The following is a list of abbreviations used in the curriculum:

AERO	-	Aerodynamics
AIM	-	Aeronautical Information Manual
AP	-	Area Planning
ARTCC	-	Air Route Traffic Control Center
ASR	-	Airport Surveillance Radar
ATC	-	Air Traffic Control
ATF	-	Aviation Training Form
ATIS	-	Automatic Terminal Information Service
ATJ	-	Aviation Training Jacket
ATS	-	Aviation Training Summary
AWOS	-	Automated Weather Observing System
BC	-	Back Course
CAI	-	Computer Aided Instruction
CDFA	-	Continuous Descent Final Approach
CIS	-	Contract Instructional Services
CNATRA	-	Chief of Naval Air Training
CO	-	Commanding Officer
CO-PC	-	Commanding Officer's Progress Check
CPT	-	Cockpit Procedures Trainer
CRIT	-	Criterion Quiz or Exam
CRM	-	Crew Resource Management
CRM-F	-	Crew Resource Management - Facilitator
CRM-I	-	Crew Resource Management - Instructor
CTS	-	Course Training Standards
DFAMFP	-	Day Familiarization Flight Procedures
DH	-	Decision Height
DINS	-	Defense Internet NOTAM System
DME	-	Distance Measuring Equipment
DP	-	Departure Procedure
EFAS	-	En route Flight Advisory Service
EMFP	-	Emergency Flight Procedures

EOB	-	End of Block
EP	-	Emergency Procedure
ESIS	-	Electronic Standby Instrument System
ET	-	Extra Training
FAA	-	Federal Aviation Administration
FAF	-	Final Approach Fix
FAR	-	Federal Aviation Regulations
FDC	-	Flight Data Center
FFL	-	Full Flap Landing
FIH	-	Flight Information Handbook
FLIP	-	Flight Information Publication
FMS	-	Flight Management System
FMS DTD	-	Flight Management System Desktop Training Device
FP	-	Flight Procedures
FRR	-	Flight Rules and Regulations
FSS	-	Flight Support Services
FTI	-	Flight Training Instruction
GCA	-	Ground-Controlled Approach
GP	-	General Planning
GPS	-	Global Positioning System
GPSFP	-	Global Positioning System Flight Procedures
HILO	-	Holding-In-Lieu-Of
IAF	-	Initial Approach Fix
IFR	-	Instrument Flight Rules
ILS	-	Instrument Landing System
IMC	-	Instrument Meteorological Conditions
IMS	-	International Military Student
IMSO	-	International Military Student Officer
INAV	-	Instrument Navigation
INS	-	Inertial Navigation System
IP	-	Instructor Pilot
IPC	-	Initial Progress Check

KIAS	-	Knots Indicated Airspeed
LECT	-	Lecture
LOA	-	Letter of Agreement
LOC	-	Localizer
LOC-BC	-	Localizer-Back Course
LPV	-	Local Performance with Vertical Guidance
LSC	-	Level Speed Change
LSO	-	Landing Signal Officer
MAP	-	Missed Approach Point
MDA	-	Minimum Descent Altitude
Metro	-	Meteorology
MFD	-	Multifunction Display
MIF	-	Maneuver Item File
MIL	-	Mediated Interactive Lecture
MPTS	-	Multi-Service Pilot Training System
NACO	-	National Aeronautical Charting Office
NATCAP	-	Naval Air Training Class Advisor Program
NATOPS	-	Naval Air Training and Operating Procedures Standardization
NAVAID	-	Navigational Aid
NDB	-	Non-directional Beacon
NFL	-	No Flap Landing
NFS	-	Naval Flight Student
NGA	-	National Geospatial Intelligence Agency
NM	-	Nautical Mile
NOTAM	-	Notices to Airmen
NSS	-	Navy Standard Score
NTAP	-	National Track Analysis Program
ODP	-	Obstacle Departure Procedure
OIS	-	Obstacle Identification Surface
OPNAV	-	Office of the Chief of Naval Operations
PAPI	-	Precision Approach Path Indicator
PAR	-	Precision Approach Radar
PAS	-	Phase Aggregate Score

PF	-	Pilot Flying
PIC	-	Pilot-in-Command
PM	-	Pilot Monitoring
PMSV	-	Pilot-to-Metro Service
P/P	-	Pen/Pencil and Paper
PT	-	Procedure Turn
PTT	-	Part Task Trainer
RAIM	-	Receiver Autonomous Integrity Monitoring
RCVA	-	Rockwell Collins Virtual Aircraft
RDO	-	Runway Duty Officer
RNAV	-	Area Navigation System
RRU	-	Ready Room UNSAT
RV	-	Radar Vectors
SDO	-	Squadron Duty Officer
SID	-	Standard Instrument Departure
SNA	-	Student Naval Aviator
SOP	-	Standard Operating Procedure
Sqdn	-	Squadron
SS	-	Self-Study
SSE	-	Simulated Single Engine
SSR	-	Special Syllabus Requirement
STAR	-	Standard Terminal Arrival
SYS	-	Systems
TAC	-	TACAN
TACAN	-	Tactical Air Navigation System
TAS	-	Traffic Avoidance System
TERPs	-	Terminal Instrument Procedures
TOLD	-	Takeoff and Landing Data
TRB	-	Training Review Board
UNSAT	-	Unsatisfactory
USN	-	United States Navy
VASI	-	Visual Approach Slope Indicator

VCOA	-	Visual Climb Over Airfield
VDA	-	Vertical Descent Angle
VDP	-	Visual Descent Point
VFR	-	Visual Flight Rules
VHF	-	Very High Frequency
VMC	-	Visual Meteorological Conditions
V_{mca}	-	Minimum Control Airspeed (air)
VNAV	-	Visual Navigation
VOR	-	VHF Omnidirectional Range
VR	-	Visual Route
V_{REF}	-	Reference Speed
V_x	-	Best Angle of Climb Speed
V_{xse}	-	Best Single Engine Angle of Climb Speed
V_y	-	Best Rate of Climb Speed
V_{yse}	-	Best Rate of Climb Speed Single Engine
WX	-	Weather
XO	-	Executive Officer

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Chapter I

General Instructions

1. Syllabus Management

a. Distribution. Participating squadron personnel.

b. Interpretation. The syllabus is directive. Should circumstances create situations not covered within the scope of this syllabus, or specific course of action appears to conflict with other directives, consult CNATRA (N71).

c. Deviations. Document all deviations on the event's ATF.

d. Changes. Recommended changes shall be submitted in accordance with CNATRAINST 1550.6G.

e. Execution. SNAs will execute all the curriculum events.

f. Syllabus Description. Intermediate E-2 MPTS consists of intermediate multi-engine training for E-2 SNAs who will go to advanced jet training. This training is flown in the T-44C aircraft. Each phase is divided into stages. Stages are grouped by like flight training regimes: Familiarization, Instrument, and Formation. Each stage is subdivided into training blocks. The training blocks consist of a specified number of flights. Maneuver item files identify the acceptable level of performance that must be achieved at the completion of each training block.

2. Training Management

a. Syllabus Progression. Fly syllabus events within each stage sequentially. Do not start a block without all prerequisites. SNAs may be in different stages or blocks simultaneously. Where applicable, SNAs will be eligible for, and shall be prepared for, more than one syllabus event. SNAs must complete all events. The flowchart on page I-3 delineates the sequence of flying events and their ground training prerequisites. System training management is designed to facilitate two graded events (flight, simulator, or exam) per SNA per day.

b. Accelerated Progression

(1) TAP (Training Acceleration Program). SNAs with previous flight experience may warrant accelerated progression through the syllabus. TAP is a specifically tailored syllabus deviation, designed/approved by the Squadron CO, printed on a Supplementary ATF, placed in the student's ATJ, specifying which syllabus events are waived based on the SNA's previous flight experience. This policy shall not be used to meet squadron production goals; it is strictly for instances where previous flight experience makes completion of all events within a block of instruction unnecessary. All ATFs for the accelerated NFS will be clearly marked "Proficiency Adv" in the appropriate ATF dropdown box." The squadron CO shall closely monitor the progress

of TAP SNAs. If performance suffers due to participation in TAP, the SNA shall return to normal syllabus progression.

(2) Proficiency Advance. SNA with demonstrated above average performance may warrant accelerated progression, known as "Proficiency Advance." Squadron CO is encouraged to support a squadron program where a SNA/SNFO may be advanced to the next block of instruction when all required items for the current block meet or exceed performance prerequisites for the follow-on block of training:

(a) The squadron CO shall closely monitor the progress of Proficiency Advanced students. If performance suffers, the CO has options to assist the NFS with meeting performance expectations, (e.g., awarding ETs).

(b) All ATFs for the Proficiency Advanced events will be marked "Proficiency Adv" in the appropriate ATF dropdown box.

(c) This policy shall not be used as a shortcut to meet squadron production goals; it is strictly for instances where demonstrated proficiency makes completion of all events within a block of instruction unnecessary.

c. Maneuver Continuity. SNAs must accomplish previously introduced maneuvers frequently enough to ensure required proficiency is maintained.

d. Hours/X (H/X). Instructor Pilots shall plan and execute missions to meet H/X as closely as practical. If actual event length varies from H/X by more than 0.3 hours, annotate reason(s) in the ATF's general comments section. An SNA's deficiency is not an acceptable reason to exceed H/X by more than 0.3 hours.

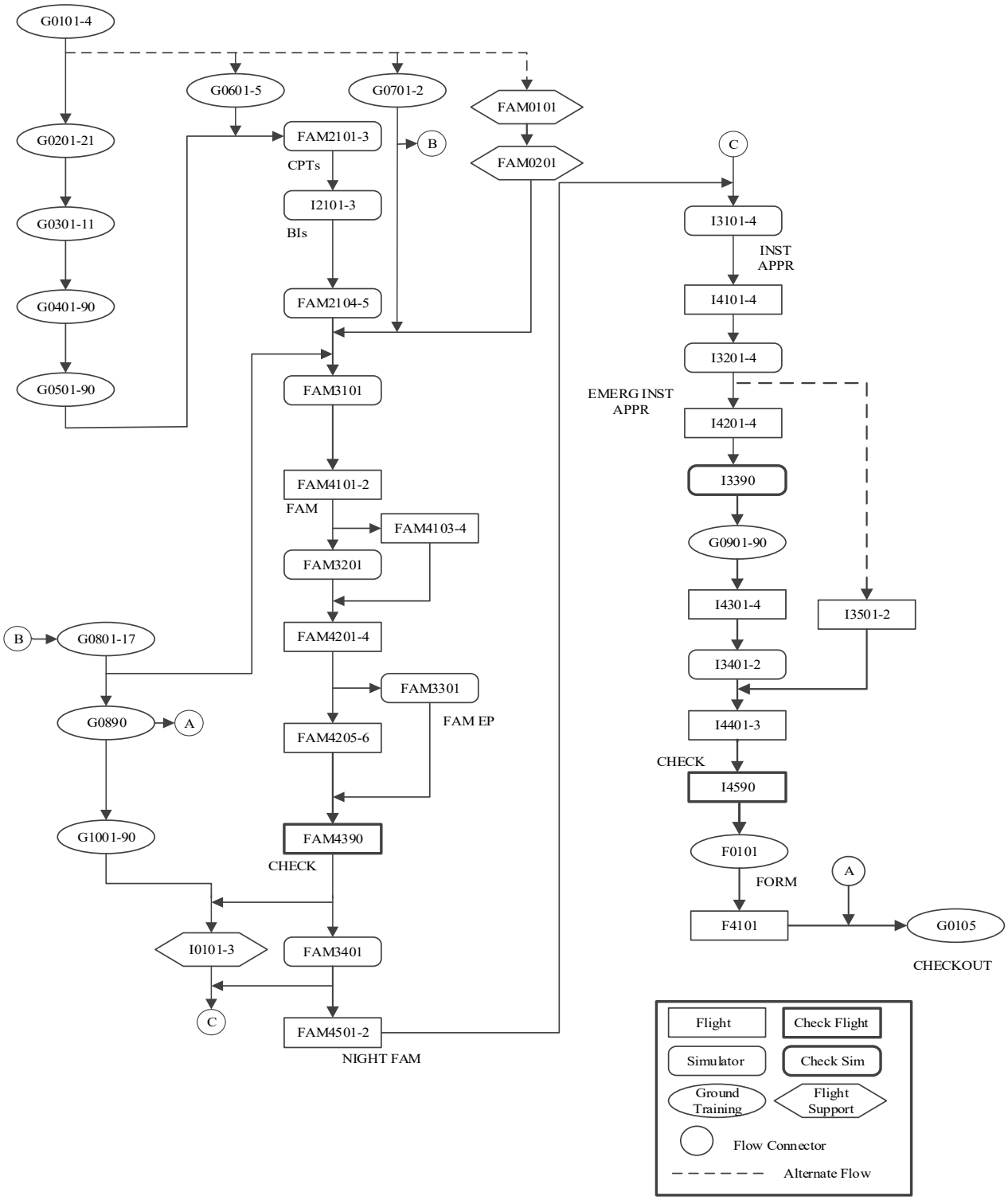
e. Special Syllabus Requirements. Unless otherwise noted, SIs may accomplish Special Syllabus Requirements (SSR) on any flight in the block. SSRs shall be completed in the specified block. Annotate completion of the SSR in the ITFs maneuver item content line and document the date on the SSR tab in T-SHARP. Assign NG/1 as the SSR maneuver grade.

f. Aviation Training Jacket Reviews

(1) The Class Advisor shall conduct weekly jacket reviews per CNATRAINST 1500.4L. ATS forms are required to be reviewed before each flight or simulator event. ATJ reviews may be made as often as warranted by the individual SNA's progress, subject to the following guidelines:

(2) All reviews shall be annotated on the CNATRA-GEN 1542/66, Jacket Review Divider, in the ATJ.

INTERMEDIATE E-2D MPTS COURSE FLOW



3. UNSAT Performance. (See also ***Progress Check***, Chapter I, paragraph 9.c.(3).

a. Flights and Simulators

(1) Event Progression. Following an UNSAT event, if a PC is not required, that event shall be repeated until the NFS satisfactorily passes the event.

(2) NFS exhibits dangerous tendencies, or progress towards meeting EOB performance guidelines is insufficient. An overall event UNSAT is at the instructor's discretion. It should be noted that an event may be graded UNSAT without any individual maneuvers graded U/2. If an overall UNSAT grade is awarded on any event other than normal syllabus flow events, the instructor shall clearly state the justification in the General Comments section of the ATF.

(3) Any UNSAT event, starting with failed academic exams in NIFE, through the final event prior to designation as a Naval Aviator or Naval Flight Officer, shall cumulatively be used in Progress Check triggers.

b. Ready Room UNSAT (RRU). A missed brief does not constitute an RRU. Missed briefs shall be documented on a Supplementary ATF and counseling or discipline shall be administered as directed by the squadron CO. An RRU is defined as either of the following:

(1) An NFS is inadequately prepared for the scheduled event. RRUs always trigger a Progress Check, so they shall always be documented on a pink version of the event's ATF. The event will be marked as incomplete with a U/2 grade in the appropriate graded item column, (e.g., General Knowledge, Procedures, etc.). Upon successful completion of the Progress Check, the original RRU event shall be flown as a take two (or greater) to complete all remaining, or appropriate items, graded as a normal event.

(2) The NFS fails a non-academic examination (NATOPS quiz, etc.).

c. Academic. Failing two exams triggers an IPC or CO-PC as appropriate.

d. Remediation

(1) A ground evaluation emphasizing the deficient areas may clear an UNSAT check flight or end of block syllabus event caused solely by ground operations.

(2) End of block UNSAT syllabus events in the Instrument stage may be cleared in the simulator if these conditions are met:

(a) The cause of the UNSAT is specific to the maneuver.

(b) The simulator is suited to the failed maneuver.

e. Restrictions. Until remediating the UNSAT:

(1) The SNA shall not fly solo.

(2) The SNA shall not accomplish any training except academic classes, examinations, and ground training events, provided the UNSAT event was not a prerequisite.

4. Training Review Board

a. The TRAWING shall conduct a TRB on all NFSs recommended for attrition.

b. Refer to 1500.4L for additional information.

5. Instructor Continuity

a. SNAs shall fly Familiarization syllabus events FAM3201, FAM4101-4 and FAM4201-2 with their on-wing.

b. Any Familiarization Stan IP may substitute as on-wing in the event the SNA's on-wing is not available and an on-wing change is not prudent.

6. Training Delays and Warmup Events Within Stage (SXX86)

a. A Warmup (WU) event is given to regain flight proficiency due to a training delay. Eligibility is based on the number of days since the last stage syllabus event, in either the aircraft or simulator. WU criteria are normally based on last event in stage because MPTS and NFOTS are built on increasing levels of skill attainment between blocks or stand-alone events. Every WU event shall ensure required skills for that stage are refreshed. WU events shall be coded as SXX86 and include a justification in the general comments section of the ATF. For more detailed information on warmup training, refer to CNATRAINST 1500.4L

(1) WUs shall be scheduled and flown as the next event. If performance warrants a WU, it shall be re-coded as the last completed dual event. If performance is sufficient, or will be sufficient by EOB, any WU may be an advancing event at IP discretion. Standard event completion criteria apply.

(2) Individual maneuver grades on WU events will not count toward calculation of NFS stage and phase grades. WU events are however eligible for overall event Unsatisfactory (UNSAT), if a significant incident not related to the stage discontinuity occurred. Note that even though stage and phase grades are not affected, UNSAT WU events do negatively affect the student's PAS and NSS since the UNSAT does count toward the total number of UNSATs (see 1500.4L, Appendix G).

(3) Ready Room UNSATs (RRU) are overall event UNSAT and count toward PC triggers. See CNATRAINST 1500.4L, paragraph 608 for amplifying information on this situation.

(4) If the break in training occurs during the transition from simulator to aircraft, a WU shall be flown in the simulator and coded as the last completed simulator in stage. The WU may be flown in the aircraft with the TRAWING Commander's approval.

(5) All WUs shall be dual for flight events or with an instructor in the simulator, except in the case of carrier qualifications, where WUs may be solo at the discretion of the respective LSO.

(6) For delays less than seven days, the specific WU criteria, promulgated in the MCG, shall apply.

(7) The instructor shall document on the ATF the reason and circumstances for awarding the WU event.

(8) Check events (SXX90) are considered part of the stage for which they are checking for WU purposes.

(9) Stage-specific EP simulators satisfy simulator WU requirements for blocks within that stage.

(10) PC WU criteria differ from normal syllabus flow WU criteria; see paragraph 703.

(11) The following table is a quick reference regarding the use of WUs with respect to stage continuity, or breaks in training:

CRITERIA FOR AWARDING WARMUP EVENTS IN STAGE		
BREAK* (DAYS)	WARMUP	REMARKS
7-13 Sim to A/C	1 Simulator	– WU is not an advancing event. – Warmup event may be flown in aircraft with the TRAWING Commander's approval.
7-13 All others	1 WU	– WU is awarded at instructor's discretion and based upon an assessment of student stage performance. – WU is prohibited if demonstrated performance is sufficient, or will be sufficient within remaining block events, by EOB.
14-30 Sim to A/C	2 Simulators	– If student performance is sufficient, second WU may be waived by squadron CO. – For blocks with a single simulator event, only one mandatory simulator WU event is required.
14-30 All others	2 WUs	– First WU may be an advancing event unless executing a safe-for-solo event. – Second WU is awarded at instructor's discretion and based upon an assessment of student stage performance. – Second WU is prohibited if demonstrated performance is sufficient, or will be sufficient within remaining block events, by EOB.

*Break = (Current Julian Date) – (Julian Date of last simulator or flight event in state).

b. Training Delays and Warmups Between Stages. WU are intended for non-curriculum breaks in training. Each curriculum is designed to allow sufficient time for academics, simulators, and flights. First flights in stage following ground training are designed and graded with the delay factored in and normally do not require a WU.

(1) Between stages a WU is required if 14 to 30 days have elapsed since any curriculum unless otherwise specified in respective MCG.

Note. Mandatory WU does not apply for students going from I21 block back to FAM21 block, (i.e., I2103 to FAM2104). The CI may grant an optional warm-up if deemed necessary.

(2) All WU events between stages shall be recorded on ATF and will include all required and relevant information to record and conduct and performance of that WU event.

c. Extended Training Delays. If the period between events is greater than 30 days, the squadron CO shall determine an appropriate WU training plan to regain NFS proficiency with the following guidance:

(1) A WU training plan should consist of a representative cross-section of events completed prior to the break in training. The WU training plan shall be placed in the NSF's ATJ.

(2) The NFS should be thoroughly briefed and provided with a copy of the WU training plan prior to execution. If the WU training plan includes academic or simulator events, the TRAWING Ground Training Officer shall be furnished with a copy of the plan to ensure the training is properly scheduled. The CO may amend the WU training plan based on NFS performance during execution of that plan.

(3) Upon successful completion of the WU training plan, the NFS shall resume normal syllabus flow.

(4) If the NFS has not regained proficiency sufficient to resume training following the WU training plan, additional WU training may be accomplished at the CO's discretion. If elected, a Supplementary ATF with an accompanying additional WU plan shall be added to the NFS's ATJ.

7. Additional Flights and Simulators

a. Extra Training Events (SXX87). All ETs shall be dual and coded as SXX87, (e.g., FAM4187). ET events include, but are not limited to:

(1) IPC or CO-PC ET Events. ETs offer the CO a powerful tool to improve NFS performance. ETs do however come with an opportunity cost of using scarce resources for individual remediation, where they would otherwise be used in normal training processes.

(2) The CO may authorize up to two ETs following an UNSAT event or up to two ETs preceding a Progress Check.

(3) International SNAs. Foreign-language Extra Training (ET) events are ***not*** authorized unless coordinated beforehand. Coordination is required between the training squadron IMSO, TRAWING IMSO and NETSAFA, with notification to CNATRA (N713) of any approved foreign-language ETs. Approval of foreign-language ETs shall be documented on a Supplementary ATF.

(4) Additional Events to Meet Minimum Syllabus Time. An event flown to meet minimum nighttime shall be flown as a FAM4587 and will meet the MIF for the block in which the ET is flown. All critical items need not be completed on this additional event.

b. Adaptation Events (SXX84). The Operations Officer may grant events required for adaptation to the flying environment when requested by the flight surgeon, (e.g., airsickness, eyeglasses, etc.).

8. Ground Training and Briefing Requirements

a. Mission Preparation, Briefings, and Debriefings

(1) EOB Events. The IP shall carefully review the Aviation Training Summary in planning the EOB event to ensure the profile includes opportunities to reach MIF on all critical items and optional items attempted in the block.

(2) Preparation. SNAs shall arrive for each flight with:

(a) Thorough knowledge of:

1. The flight's Discuss Items, as listed in Chapters IV-VII.

2. Procedural knowledge of all items for the event's training block.

(b) A flight profile tailored to training requirements, weak areas, and continuity.

(c) The latest ATS for the stage.

(3) Briefing. Thoroughly cover the mission's:

(a) Event Discuss Items, as listed in Chapters IV-VII.

(b) Specific objectives.

(c) Techniques and required procedures for accomplishing those objectives.

(d) Planned profile and contingencies.

(4) Debriefing

(a) After each event, the instructor shall critique the SNA's performance using cause and effect analysis, particularly with respect to the CTS.

(b) The mission's complexity and SNA's progress will govern the time required for the debrief.

b. Emergency Procedures (EP) Briefing and Training

(1) EP training builds the SNA's confidence in the aircraft. The IP shall conduct EP training on all dual aircraft events, either on the ground or in the aircraft. Correct procedural deficiencies through additional instruction and study assignments.

(2) Incorporate EP training into simulator events when practical; however, instructional block objectives take precedence.

(3) Grade the SNA's overall EP knowledge and performance under Emergency Procedures.

9. Mission Grading Procedures and Evaluation Policies

a. General Grading and Evaluation Policy. Maneuver item files listed in the MPTS are minimum stage or phase completion standards per maneuver.

b. Grading Procedures (Aircraft and Training Devices)

(1) Absolute Maneuver Grading. Use the following grading scale to document the SNA's characteristic performance on maneuvers attempted during each dual event. This grading scale is an absolute grading scale. Judge the SNA's proficiency ***only*** against the item's course training standard. (See ***Student Performance Measurement and Application of Standards,*** page x, Course Data, paragraph 21.).

(a) Demonstrated or Optional (NG/1).

1. When the IP demonstrates the maneuver and the SNA does not subsequently perform it during the event.

2. When the student performs a maneuver that is optional in the block.

3. To indicate accomplishing all SSRs for that event. Specify the completed SSR in the ATF's maneuver item content line and document date of exposure on the SSR tab.

(b) Unable (U/2 Level). Performance is unsafe or lacks sufficient knowledge, skill, or ability. Deviations greatly exceed CTS, significantly disrupting performance. Corrections significantly lag deviations or aggravate the deviation.

(c) Fair (F/3 Level). Performance is safe, but with limited proficiency. Deviations exceed CTS, detracting from performance. Corrections noticeably lag deviations, and may not be appropriate.

(d) Good (G/4 Level). Performance meets or positively exceeds CTS. Deviations outside CTS are allowed, provided they are brief, minor, and do not affect safety of flight. Corrections must be appropriate and timely.

(e) Excellent (E/5 Level). Greatly surpasses CTS. Performance is correct, efficient, and skillful. Deviations are very minor. Corrections, if required, are initiated by the SNA and are appropriate, smooth, and rapid.

(2) Overall Event Grades. Overall event grades represent the SNA's progression through MPTS. Grade events "Pass or" or "UNSAT." Use the following definitions to characterize event grades.

(a) Pass

1. Prior to EOB. Progress is adequate to meet standards by EOB.

2. EOB. The SNA's performance meets or exceeds standards.

(b) UNSAT. The SNA exhibits dangerous tendencies, or progress towards meeting EOB standards is insufficient. UNSAT overall is at the instructor's discretion. It should be noted that an event may be graded UNSAT without any individual maneuvers graded 2/Unable. Document all UNSAT events on a pick ATF.

(3) Awarding Overall Event Grades. The SNA's overall grade is based on the SNA's performance against the MIF for that event. The following rules govern overall event grading:

(a) EOB. Performance must meet MIF by EOB. As long as MIF was met in block at least two times, a NFS can progress past EOB if a maneuver does not meet MIF on EOB event.

(4) Maneuver Requirements. For each block:

(a) Mandatory Items. Items with a number and a plus (+) are mandatory and the SNA must meet the required proficiency by EOB. As long as MIF was met in block at least two times, a NFS can progress past EOB if a maneuver does not meet MIF on EOB event.

(b) Optional Items. Items with a MIF of 1 are optional; however, if flown, can only be graded as a NG/1 or U/2.

(c) Not Demonstrated or Not Performed. The IP will not demonstrate, nor will the SNA perform:

1. Unnumbered items.

2. Items not in the stage.

3. Exceptions:

a. Weather-driven instrument approaches.

b. Prebriefed maneuvers for IP proficiency.

(5) Incomplete Events. In general, IPs should consider an event complete if able to accomplish the requirements in paragraph (a) below. This is particularly true when weather precludes accomplishing certain maneuver items, but the IP is able to emphasize training on other maneuver items. Subsequent events in the block, when available, can reverse this emphasis, hence achieving overall training balance. If an SNA has had ample opportunity to learn a task and subsequently flies a short mission, do not incomplete the mission solely to provide unwarranted extra training.

(a) Assessment. Assess the event complete if:

1. A significant portion of the sortie was used for student training, and

2. Sufficient events remain in the block to redress the imbalance, and

3. The SNA passes the event.

4. Otherwise, assess the event incomplete.

(b) Completion Events

1. An event may both complete a previous event and count as an Advancing X.

2. For events flown exclusively to clear an incomplete, grades on maneuvers repeated from the incomplete event do not count towards the SNA's score, except where the grade assigned for the repeated item is lower that the lowest grade previously assigned on that item from all previous attempts on that item.

(c) Simulator Event Completion. Assess a simulator event complete if the SNA has received a full training period per the master curriculum guide.

c. Policies for Evaluation Flights and Ground Evaluations

(1) Authorized Evaluators. The squadron CO will designate check pilots for each stage.

(2) Check Flights (SXX90)

(a) Single Event Training Blocks. Check flights amount to single event training blocks.

1. Should fly a representative cross section of optional maneuvers.

2. Check pilots may allow SNAs to reattempt maneuvers.

3. The entire event should be devoted to assessing the SNA's ability and readiness to progress to the next stage of training. All maneuvers indicated with a plus (+) are check flight critical and must be accomplished to MIF.

4. The SNA should be able to demonstrate required levels of proficiency without instructor assistance; however, instruction is allowed on check flights and SNAs may reattempt maneuvers at the check pilot's discretion.

(b) Incomplete Check Flight. The check flight shall be incomplete when:

1. Any (+) item was not flown, or

2. The check pilot was unable to sample sufficient examples of a given maneuver to assess the SNA's overall performance, or

3. The flight is UNSAT.

Note: The subsequent flight need only include maneuvers required to complete the check if the previous flight was established as a "Pass".

(c) Instrument or Navigation Stage Check Failure. If the SNA fails an Instrument or Navigation stage check because of an UNSAT pattern or landing not directly related to the stage being evaluated, any subsequent ET event may be flown as a familiarization event.

(3) Progress Check

(a) The Progress Checks flown in the aircraft or simulator are holistic reviews of SNA's proficiency, judgment, air sense, and overall ability to maneuver the aircraft safely and confidently. The intent of every Progress Check is to determine whether the SNA has the potential to reach the defined training standards of his or her current phase of training within the designated TTT, while demonstrating the potential to successfully complete advanced and FRS-level training process.

(b) Refer to 1500.4L for additional information.

10. Special Instructions and Restrictions

a. Flight Hour and Event Requirements and Restrictions

(1) Programmed Hours and Events. Programmed syllabus flight hours are listed on page ix. Event lengths, SXX86, 87, 88, and 89 events will cause variation. Accomplish all syllabus events.

(2) Minimum Night Hours. 4.0 hours.

(3) Maximum Daily SNA Activities (Aircraft, Simulator, or Academic). SNAs shall not exceed two graded activities during one duty day.

(4) Minimum SNA Turn-Times. One hour is required between debriefing of an event and the brief for a follow-on or simulator event. This does not apply to out-and-in, cross-country, or safe-for-solo to hot-seat profiles; however, the instructor shall ensure adequate debrief and brief time is allocated.

(5) Crew Day. The period from the beginning of the SNA's first event or official duty of the day until the completion of the last event of the day, including associated paperwork and debrief. Crew day shall not exceed 12 hours.

(6) Crew Rest. A minimum of 12 hours shall elapse between the conclusion of the NFS's last scheduled event of the day (including associated debrief) and the NFS's first scheduled event (including associated brief) of the following day. For scheduled non-graded events (classes, CAI, MIL, watch-standing duty days, this period may be shortened to eight hours. After six consecutive scheduled days, the NFS shall receive one day off. Official duty, squadron training, or standby scheduling do not qualify as a day off. No training may be accomplished on a duty day.

b. Aircraft and Simulator Interchangeability. Simulator events may be substituted in the aircraft when the simulator is unavailable for extended periods of time.

Chapter II

Ground Training

Blk #	Media	Title	Events	Hrs	Blk Name
G01	Sqdn/ Lect	Indoctrination	5	4.1	ASI

1. Prerequisites

   a. G0101 prior to G0102-4 (in order).

   b. G0890 (Instrument Navigation Exam) and F4101 prior to G0105.

2. Events

G0101	Lect	Squadron Welcome Aboard	1.0
G0102	Sqdn	Squadron Policies	1.0
G0103	Sqdn	Facilities Tour	1.0
G0104	Lect	Academic Indoctrination (Class)	1.0
G0105	Admin	Checkout	0.1

3. Syllabus Notes

   a. G0104 is conducted in BLDG 1824.

   b. G0105 is an administrative event and allows a final check that all requirements for completion of syllabus have been met.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G02	Class	Systems I	22	22.0	SYS

1. Prerequisites

   a. G0104 Academic Indoctrination prior to G0201-19 (any order).

   b. G0219 prior to G0220-21 (in order).

2. Events

G0201	MIL	Introduction to Aircraft Systems	0.5
G0202	CAI	T-44C General Aircraft I	0.5
G0203	MIL	T-44C General Aircraft II	0.5
G0204	CAI	T-44C Electrical Systems I	1.0
G0205	MIL	T-44C Electrical Systems II	1.5
G0206	CAI	T-44C Power Plant and Related Systems I	2.0
G0207	MIL	T-44C Power Plant and Related Systems II	1.0
G0208	CAI	T-44C Propeller System I	1.0
G0209	MIL	T-44C Propeller System II	1.0
G0210	CAI	T-44C Fuel System I	1.0
G0211	MIL	T-44C Fuel System II	1.0
G0212	CAI	T-44C Flight Control System I	0.5
G0213	MIL	T-44C Flight Control System II	0.5
G0214	CAI	T-44C Landing Gear System I	1.0
G0215	MIL	T-44C Landing Gear System II	1.0
G0216	Sqdn	T-44C Aircraft Tour	1.0
G0217	Lect	T-44C Simulator Brief	0.5
G0218	CAI	T-44C Environmental Systems I	1.5

2. Events (cont.)

G0219	MIL	T-44C Environmental Systems II	1.5
G0220	Lect	T-44C Course Review	1.5
G0290	CAI Test	T-44C Aircraft Systems I Exam	1.5
G0221	Lect	T-44C Aircraft Systems I Exam Review	0.5

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G03	Class	Systems II	11	9.5	SYS

1. Prerequisites. G02 block prior to G0301-11 (any order).

2. Events

G0301	CAI	T-44C Flight Instruments I	0.5
G0302	MIL	T-44C Flight Instruments II	1.0
G0303	CAI	T-44C Navigation and Communication I	1.0
G0304	MIL	T-44C Navigation and Communication II	1.0
G0305	CAI	T-44C Autopilot System I	1.0
G0306	MIL	T-44C Autopilot System II	1.0
G0307	CAI	T-44C Multi-Function Display I	1.0
G0308	MIL	T-44C Multi-Function Display II	0.5
G0309	MIL	T-44C Flight Management System I	1.0
G0310	Lect	T-44C Flight Management System II	0.5
G0311	PTT/ Lab	Flight Management System Practical Application	1.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G04	Class	Meteorology	4	11.0	METRO

1. Prerequisites. G03 block prior to G0401-90 (in order).

2. Events

G0401	CAI	Meteorology I	4.0
G0402	MIL	Meteorology II	3.0
G0403	MIL	Meteorology Review	2.0
G0490	CAI Test	Meteorology Exam	2.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G05	Class	Aerodynamics	8	24.5	AERO

1. Prerequisites

   a. G0490 prior to G0501-2 (any order).

   b. G0502 prior to G0503-7 (any order), prior to G0590.

2. Events

G0501	CAI	T-44C Aerodynamics/Power Performance	4.0
G0502	Lect	Introduction to T-44 Aerodynamics	5.0
G0503	Lect	T-44 Aerodynamics Lecture (Weight and Balance)	2.5
G0504	Lect	T-44 Aerodynamics Lecture	2.5
G0505	Lab	Aerodynamics Lab (Weight and Balance Problem)	3.0
G0506	Lab	T-44 Aerodynamics Lab (TOLD Problem)	3.0
G0507	Lect	Aerodynamics Review	0.5
G0590	P/P	T-44C Aerodynamics Exam	4.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G06	Class/ Exam	Flight Procedures	5	9.25	See Below

1. Prerequisites

   a. G0104 prior to G0601-5 (any order).

   b. Academic instructors determine order of completion.

2. Events

G0601	P/P Exams	T-44C CRIT Exams	2.0	FP
G0602	CAI	T-44C Familiarization Flight Procedures	2.0	EMFP
G0603	Lect	T-44C Flight Procedures Lecture	2.0	DFAMFP
G0604	Lect	Course Rules Lecture	2.5	FP
G0605	Lect	Flight-Line Driver's License Brief	.75	FP

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G07	Class	Crew Resource Management	2	3.0	CRM

1. Prerequisites. G0104 prior to G0701-2 (any order).

2. Events

G0701	MIL	Seven CRM Skills	2.0
G0702	Lect	Sensory Problems/Spatial Disorientation	1.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Event	Hrs	Blk Name
G08	Class	Instrument Flight Rules - Phase I	18	25.5	IFR

1. Prerequisites

   a. G07 block prior to G0801-17 (any order).

   b. G0817 prior to G0890.

   c. G0890 prior to G0105 (Checkout).

2. Events

G0801	Lect	Introduction to Inst. Flight Rules	0.5
G0802	Lect	Introduction to FLIP Publications	0.5
G0803	CAI	Instrument Approaches I	3.5
G0804	MIL	Navigational Aids Lecture	1.0
G0805	MIL	GNSS/RNP/RNAV Lecture	1.0
G0806	MIL	Airspace Lecture	1.0
G0807	MIL	Communications	0.5
G0808	MIL	Clearances Lecture	1.0
G0809	MIL	Departures Lecture	1.0
G0810	MIL	En route Procedures Lecture	1.0
G0811	MIL	Holding Lecture	1.0
G0812	MIL	EKB Introduction	1.0
G0813	CAI	Flight Planning	3.5
G0814	MIL	Terminal Procedures Lecture	1.0

2. Events (cont.)

G0815	MIL	Instrument Approaches II	1.0
G0816	P/P	INAV Practice Exam	2.0
G0817	Lect	INAV Review	3.0
G0890	CAI Test	INAV Instrument Exam	2.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
G09	Class	Instrument Flight Rules - Phase II	7	21.5	IFR

1. Prerequisites

   a. I3390 prior to G0901-2.

   b. G0902 prior to G0903-4.

   c. G0901 and G0903-5 (any order) prior to G0906.

   d. G0906 prior to G0990.

2. Events

G0901	Lect	Introduction to Advanced Flight Planning	1.0
G0902	Lect	Navigation Computer and Multi-Engine Fuel Log Practice Problems	2.5
G0903	Lect	Completion of the DD-1801 Flight Plan	1.0
G0904	SS	Practice flight Planning Problem	8.5
G0905	Lect	Intro to EKB Flight Planning	2.0
G0906	MIL	IFR Final Exam Review	2.5
G0990	P/P Exam	IFR Final Exam	4.0

3. Syllabus Notes. None.

4. Discuss Items. None

Blk #	Media	Title	Event	Hrs	Blk Name
G10	Class	Flight Rules and Regulations (FRR)	7	13.5	FRR

1. Prerequisites. G0801-17 prior to G1001-90.

2. Events

G1001	Lect	Flight Rules and Regulations (FRR) Lecture	2.0
G1002	CAI	FRR	3.0
G1003	P/P	FRR Practice Exam	2.0
G1004	MIL	FRR Review	2.0
G1005	Lect	GNSS/RNP/RNAV Flight Planning Lecture	1.0
G1006	PTT	Flight Management System Practical Application	1.5
G1090	CAI Test	FFR Exam	2.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Chapter III

NATOPS Training

This chapter does not apply to Intermediate E-2 MPTS.

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Chapter IV

Familiarization Training

Blk #	Media	Title	Events	Hrs	Blk Name
FAM01	PTT	Checklist Orientation	1	2.0	CKLST

1. Prerequisites. G0104 (Academic Indoctrination).

2. Events

FAM0101	PTT	Checklist Orientation	2.0

3. Syllabus Notes

a. FAM0101 is a self-paced event where the student will use the Checklist Part Task Trainer to practice, at a minimum, the following checklists: Before start, after start, before takeoff, after landing, and secure.

b. Although not required, FAM2105 will normally be completed prior to FAM0101.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	Blk Name
FAM02	Flight Line	Familiarization Brief	1	5.0	DFAMFP

1. Prerequisites. FAM0101 (Checklist Orientation).

2. Events

FAM0201	Flight Line	Familiarization Brief	5.0

3. Syllabus Notes. FAM0201 has no required location, but the briefing should include a visit to Base Ops and Weather Shop, Aircraft Issue, the squadron briefing spaces, and the aircraft.

4. Discuss Items. CRM – seven skills and callouts, threat and error management, FTI and NATOPS manual use (verify changes posted), local operations, flight schedule, PIC CRM, observer duties, safety and standardization programs, weight and balance, performance charts, go/no-go criteria, Training-Time-Out policy, ditching, forced landing, area and course rules familiarization, FAA Letter of Agreement, squadron SOP, squadron standardization notes, Wing SOP, TAS operation, headset operation, personal and emergency equipment, aircraft interior and exterior inspection, emergency egress procedures, SDO indoctrination, and oxygen system operation and requirements.

Blk #	Media	Title	Events	Hrs	H/X
FAM21	UTD/OFT	Cockpit Procedures	5	7.5	1.5

1. Prerequisites

a. G0290 (T-44C Aircraft Systems Exam).

b. G0490 (Meteorology Exam).

c. G0590 (T-44C Aerodynamics Exam).

d. All G06 events (Flight Procedures block).

e. I2103 must be completed prior to FAM2104.

2. Syllabus Notes

a. Practice all checklists, applicable FTI briefings, radio calls, and basic aircraft control. Ensure NFS's checklist proficiency is adequate to proceed to flight operations.

b. Multiple items are listed as discuss items. Due to time constraints, it will not be possible to discuss all items prior to the simulator event (SIM), therefore, a discuss item may be addressed during or prior to the SIM.

c. During all FAM21blk events, each normal checklist should be performed if it has been previously introduced or discussed.

d. FAM2104 shall be conducted in the OFT.

3. Special Syllabus Requirements

FAM2101
Perform RAIM check.

FAM2103
Mechanical stops in throttle quadrant.

FAM2104
V_{mca} demo.

4. Discuss Items

FAM2101
Seat and rudder pedal adjustment, fuel management panel, pilot instrument panel, engine instruments and switches, center instrument panel, annunciator panel analysis, right seat instrument panel, circuit breaker panels, control pedestal, overhead control panel, MFD, checklists: before start, engine start, after start, brake check (out of chocks), engine run-up, takeoff, after landing, and secure.

FAM2102
Takeoff procedures, checklist management, landing procedures, checklists: climb, cruise, descent, approach, landing, start malfunctions, and MFD operations, CRM callouts, and pressurization system malfunctions.

FAM2103
Loss of brakes, hot brakes, brake fire, mechanical stops in throttle quadrant, inadvertent condition lever to fuel cutoff, engine failure at altitude, anti-ice and deice systems autopilot disengagement, and ground emergencies, anti-ice and deice systems, and windshield heating failure.

FAM2104
Vmca demo, dynamic engine cut, engine failure or fire during or after takeoff, fuel system malfunctions, smoke or fume emergencies, in-flight damage, bird strikes, wing or uncontrollable fire, ditching.

FAM2105
Flap malfunctions, no-flap landings, gear malfunctions, unsafe gear and gear up landings, engine malfunctions, propeller malfunctions, electrical malfunctions, and air-starts.

5. Block MIF

CTS REF	MANEUVER	FAM2105
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
3	Start Malfunctions	4+
6	Headwork/Situational Awareness	2+
7	Basic Air Work	3+
8	Mission Planning/Briefing/Debriefing	3+

MIF continued on next page.

CTS REF	MANEUVER	FAM2105
9	Ground Operations	3+
10	Takeoff	3+
11	Aborted Takeoff	2+
13	Dynamic Engine Cut	1
20	Power-on Ditch	2+
22	Cockpit Procedures	4+
23	Radio Communications	2+
31	Pilot Flying/CRM	2+
32	Pilot Monitoring/CRM	1
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
FAM31	OFT	Familiarization	1	1.5	1.5

1. Prerequisites

a. FAM2105 (Cockpit Procedures).

b. G07 block (CRM).

c. FAM0201 (Familiarization Brief).

2. Syllabus Notes

a. This event should concentrate on basic air work, high-work maneuvers, landing patterns, and checklist management.

b. CI will instruct event.

3. Special Syllabus Requirements. None.

4. Discuss Items. Basic air work, high-work maneuvers, landing patterns, and checklist management.

5. Block MIF

CTS REF	MANEUVER	FAM3101
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
3	Start Malfunctions	1
6	Headwork/Situational Awareness	2+
7	Basic Air Work	2+
8	Mission Planning/Briefing/ Debriefing	3+
9	Ground Operations	3+
10	Takeoff	3+
11	Aborted Takeoff	1

MIF continued on next page.

CTS REF	MANEUVER	FAM3101
12	Departure	3+
13	Dynamic Engine Cut	1
14	Level Speed Change	3+
15	Turn Pattern	3+
16	Slow Flight	3+
17	Approach to Stall	2+
18	SSE Wave-off at Altitude	1
19	Emergency Descent	1
20	Power-on Ditch	1
20	SSE Ditch	1
20	Power-off Ditch	1
21	In-Flight Planning	3+
22	Cockpit Procedures	4+
23	Radio Communications	2+
24	Overhead/Break Entry	2+
25	Course Rules	3+
26	Wave-off	2+
26	SSE Wave-off	1
27	Landing Pattern	2+
27	No-Flap Pattern	2+
27	SSE Landing Pattern	1
28	Landing	2+
28	NFL	2+
28	FFL	2+
28	SSE Landing	1
29	Touch-and-Go	2+
31	Pilot Flying/CRM	2+
32	Pilot Monitoring/CRM	2+
33	Clearing	3+

Blk #	Media	Title	Events	Hrs	H/X
FAM41	T-44C	Familiarization	4	8.0	2.0

1. Prerequisites. FAM3101 (Familiarization).

2. Syllabus Notes

a. This block should concentrate on basic air work, high work maneuvers, landing patterns, and checklist management.

b. On-wing flights will be FAM4101-4.

c. On-wing SNAs shall taxi aircraft for all events in this block.

d. On a flight where a system is briefed, the SNA shall be given a simulated malfunction associated with that system.

e. Complete a minimum of two passes in landing pattern per event with IP as PF and SNA as PM.

3. Special Syllabus Requirements

FAM4101
Oxygen mask familiarization and utilization.

4. Discuss Items

FAM4101
Outside scan techniques, see and avoid, takeoff, crosswind takeoff and landing, aborted takeoff, landing pattern, full-stop landings, touch-and-go procedures, training-time-out, NATOPS brief, dual engine wave-offs, aircraft engine operating limits, engine start procedures, abnormal starts and malfunctions, and brake system and malfunctions.

FAM4102
Porpoised landings, aircraft airframe operating limits, full-flap landings, engine fire on deck, MFD operation, stalls and spin recovery, no-flap landings, ditching power-on, and fuel system and malfunctions.

FAM4103
Lost communications (VFR), smoke and fire of unknown origin, oxygen system use, emergency descent, smoke and fume elimination, and electrical system and malfunctions.

FAM4104
Engine system and malfunctions, engine failure during take-off, right-hand pattern, and night or IMC ditch versus day VMC ditch.

5. Block MIF

CTS REF	MANEUVER	FAM4104
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
3	Start Malfunctions	4+
6	Headwork/Situational Awareness	3+
7	Basic Air Work	3+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	3+
10	Takeoff	3+
11	Aborted Takeoff	3+
12	Departure	3+
13	Dynamic Engine Cut	2+
14	Level Speed Change	4+
15	Turn Pattern	4+
16	Slow Flight	4+
17	Approach to Stall	4+
18	SSE Wave-off at Altitude	2+
19	Emergency Descent	3+
20	Power-on Ditch	3+
20	SSE Ditch	3+
20	Power-off Ditch	3+
21	In-Flight Planning	3+

MIF continued on next page.

CTS REF	MANEUVER	FAM4104
22	Cockpit Procedures	4+
23	Radio Communications	3+
24	Overhead/Break Entry	3+
25	Course Rules	3+
26	Wave-off	3+
26	SSE Wave-off	1
27	Landing Pattern	4+
27	No-Flap Pattern	3+
27	SSE Landing Pattern	2+
28	Landing	3+
28	NFL	3+
28	FFL	3+
28	SSE Landing	2+
29	Touch-and-Go	3+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
FAM32	OFT	Emergency Familiarization	1	1.5	1.5

1. Prerequisites. FAM4102 (Familiarization).

2. Syllabus Notes. Event shall be flown with on-wing or Familiarization Stan IP in the right seat.

3. Special Syllabus Requirements. Complete manual gear extension.

4. Discuss Items. Emergency procedures, simulated emergency scenarios, CRM during simulated emergencies, dynamic engine cut, SSE pattern work, and SSE (landings, wave-offs, touch-and-go), landing gear system and malfunctions.

5. Block MIF

CTS REF	MANEUVER	FAM3201
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
3	Start Malfunctions	4+
4	Windmilling Air-Start	1
5	Starter-Assisted Air-Start	1
6	Headwork/Situational Awareness	3+
7	Basic Air Work	3+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	3+
10	Takeoff	3+
11	Aborted Takeoff	3+
12	Departure	3+
13	Dynamic Engine Cut	2+
15	Turn Pattern	1
16	Slow Flight	1
17	Approach to Stall	1
18	SSE Wave-off at Altitude	1

MIF continued on next page.

CTS REF	MANEUVER	FAM3201
19	Emergency Descent	1
20	Power-on Ditch	1
20	SSE Ditch	1
20	Power-off Ditch	1
21	In-Flight Planning	3+
22	Cockpit Procedures	4+
23	Radio Communications	3+
24	Overhead/Break Entry	3+
25	Course Rules	1
26	Wave-off	3+
26	SSE Wave-off	2+
27	Landing Pattern	4+
27	No-Flap Pattern	3+
27	SSE Landing Pattern	2+
28	Landing	3+
28	NFL	3+
28	FFL	3+
28	SSE Landing	2+
29	Touch-and-Go	3+
30	SSE Full-Stop	1
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
FAM42	T-44C	Emergency Familiarization	6	10.0	See Syllabus Notes c. and d.

1. Prerequisites

a. FAM3201 (Emergency Familiarization).

b. FAM4104 (Familiarization).

2. Syllabus Notes

a. The purpose of this block is to continue basic air work while introducing additional emergency procedures, specifically SSE work.

b. CRM should be emphasized during all flights, especially during SSE training.

c. FAM4201-2 – 2.0 hours per event.

d. FAM4203-6 – 1.5 hours per event.

e. On-wing flights will be FAM4201-2.

f. Windmilling air-start shall be accomplished once during the FAM42 block and annotated in the comments section of the ATF.

g. Starter-assisted air-start shall be accomplished once during the FAM42 block and annotated in the comments section of the ATF.

h. SSE full-stop shall be accomplished and graded during this block following IP demonstration.

i. Complete a minimum of two passes in landing pattern per event with IP as PF and SNA as PM.

j. Where a system is briefed, the SNA should be given a simulated malfunction associated with that system.

3. Special Syllabus Requirements

FAM4201
SSE full-stop demo.

4. Discuss Items

FAM4201
Actual versus simulated engine shutdown, engine secure and restart, PIC CRM during SSE, P-factor, starter-assisted air-start, SSE pattern work, SSE (landings, wave-offs, touch-and-go), and SSE full-stop.

FAM4202
Simulated dual engine failure, ditching – power-off, windmilling air-start, landing gear manual extension, and propeller system and malfunctions.

FAM4203
Engine fire in flight, anti-icing system, and flight control system and malfunctions.

FAM4204
Aircraft operating limits, forced landing, fuel system, and environmental and pressurization system.

FAM4205
Engine system and electrical system.

FAM4206
Inadvertent IMC, right seat positioning, pilot monitoring and right seat responsibilities, and propeller system.

5. Block MIF

CTS REF	MANEUVER	FAM4206
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
3	Start Malfunctions	4+

MIF continued on next page.

CTS REF	MANEUVER	FAM4206
4	Windmilling Air-Start	3+
5	Starter-Assisted Air-Start	3+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	4+
10	Takeoff	4+
11	Aborted Takeoff	4+
12	Departure	4+
13	Dynamic Engine Cut	4+
15	Turn Pattern	1
16	Slow Flight	1
17	Approach to Stall	4+
18	SSE Wave-off at Altitude	1
19	Emergency Descent	3+
20	Power-on Ditch	4+
20	SSE Ditch	4+
20	Power-off Ditch	4+
21	In-Flight Planning	4+
22	Cockpit Procedures	4+
23	Radio Communications	3+
24	Overhead/Break Entry	4+
25	Course Rules	4+
26	Wave-off	4+
26	SSE Wave-off	4+
27	Landing Pattern	4+
27	No-Flap Pattern	4+
27	SSE Landing Pattern	4+
28	Landing	4+
28	NFL	4+

MIF continued on next page.

CTS REF	MANEUVER	FAM4206
28	FFL	3+
28	SSE Landing	4+
29	Touch-and-Go	4+
30	SSE Full-Stop	3+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
FAM33	OFT	Familiarization Emergency Procedures/CRM	1	1.5	1.5

1. Prerequisites. FAM4204 (Emergency Familiarization).

2. Syllabus Notes

a. At least one presentation of malfunctions for each major system should be used. In addition, instructors are encouraged to give SNAs simulated emergencies during familiarization maneuvers to improve basic aircraft control.

b. Emphasis will be placed on procedural knowledge, judgment, and CRM skills. With two SNAs in the simulator, PM shall be evaluated on radio communications and CRM.

c. Communications with simulated seagull, approach control, or tower are required.

d. SNAs shall follow all EP scenarios to a logical conclusion, unless the IP resets aircraft conditions.

e. CI will instruct event.

3. Special Syllabus Requirements. None.

4. Discuss Items. Maintaining aircraft control during emergencies, pattern considerations, communications, and CRM.

5. Block MIF

CTS REF	MANEUVER	FAM3301
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
3	Start Malfunctions	1
6	Headwork/Situational Awareness	4+
7	Basic Air Work	3+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	4+

MIF continued on next page.

CTS REF	MANEUVER	FAM3301
10	Takeoff	4+
11	Aborted Takeoff	4+
12	Departure	4+
13	Dynamic Engine Cut	1
14	Level Speed Change	1
15	Turn Pattern	1
16	Slow Flight	1
17	Approach to Stall	1
19	Emergency Descent	1
20	Power-on Ditch	1
20	SSE Ditch	1
20	Power-off Ditch	1
22	Cockpit Procedures	4+
23	Radio Communications	1
26	Wave-off	1
26	SSE Wave-off	1
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	1

Blk #	Media	Title	Events	Hrs	H/X
FAM43	T-44C	Familiarization Check Flight	1	1.5	1.5

1. Prerequisites

a. FAM4206 (Emergency Familiarization).

b. FAM3301 (Familiarization EP/CRM).

2. Syllabus Notes

a. This event will be an evaluation of Familiarization and VFR flying involving a representative cross section of maneuvers previously presented.

b. At the discretion of IP, pilot monitoring and right seat responsibilities during three IP landings may be re-accomplished if break (Julian date – Julian date last flown) is greater than three days between FAM4206 and FAM4390.

3. Special Syllabus Requirements. None.

4. Discuss Items. CRM, pilot monitoring responsibilities, and touch-and-go abort decisions. Operation limits quiz and RDO briefing.

5. Block MIF

CTS REF	MANEUVER	FAM4390
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
3	Start Malfunctions	1
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	1
10	Takeoff	4+
11	Aborted Takeoff	1
12	Departure	1

MIF continued on next page.

CTS REF	MANEUVER	FAM4390
13	Dynamic Engine Cut	4+
14	Level Speed Change	1
15	Turn Pattern	1
16	Slow Flight	1
17	Approach to Stall	4+
18	SSE Wave-off at Altitude	1
20	Power-on Ditch	1
20	SSE Ditch	1
20	Power-off Ditch	1
21	In-Flight Planning	4+
22	Cockpit Procedures	4+
23	Radio Communications	3+
24	Overhead/Break Entry	1
25	Course Rules	1
26	Wave-off	4+
26	SSE Wave-off	4+
27	Landing Pattern	4+
27	No-Flap Pattern	4+
27	SSE Landing Pattern	4+
28	Landing	4+
28	NFL	4+
28	FFL	1
28	SSE Landing	4+
29	Touch-and-Go	4+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+

Blk #	Media	Title	Events	Hrs	H/X
FAM34	OFT	Night Familiarization	1	1.5	1.5

1. Prerequisites. FAM4390 (Familiarization Check Flight).

2. Syllabus Notes. CI will instruct event.

3. Special Syllabus Requirements. None.

4. Discuss Items. Night flying environment, field lighting, aircraft lighting, observer duties, outside scan techniques, see and avoid, porpoised landing, CRM guidelines and callouts, and night instrument approaches.

5. Block MIF

CTS REF	MANEUVER	FAM3401
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	3+
7	Basic Air Work	3+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	4+
10	Takeoff	3+
11	Aborted Takeoff	3+
12	Departure	3+
22	Cockpit Procedures	4+
23	Radio Communications	3+
24	Overhead/Break Entry	3+
25	Course Rules	1
26	Wave-off	3+
26	SSE Wave-off	3+
27	Landing Pattern	3+

MIF continued on next page.

CTS REF	MANEUVER	FAM3401
27	No-Flap Pattern	3+
27	SSE Landing Pattern	3+
28	Landing	3+
28	NFL	3+
28	FFL	3+
28	SSE Landing	3+
29	Touch-and-Go	3+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+

Blk #	Media	Title	Events	Hrs	H/X
FAM45	T-44C	Night Familiarization	2	3.0	1.5

1. Prerequisites. FAM3401 (Night Familiarization).

2. Syllabus Notes. An instrument approach shall be demonstrated by the IP on each event with SNA as PM, emphasizing CRM callouts and radio communications.

3. Special Syllabus Requirements

FAM4501
Visual approach demonstration.

4. Discuss Items

FAM4501
Local operations, right-hand pattern and CRM night or IMC ditching, runway/taxiway/parking signs and symbols, runway markings, and night visual approaches.

FAM4502
Aircraft operating limits, CRM guidelines/callouts.

5. Block MIF

CTS REF	MANEUVER	FAM4502
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
3	Start Malfunctions	1
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	3+
9	Ground Operations	4+
10	Takeoff	4+
11	Aborted Takeoff	4+
12	Departure	4+
22	Cockpit Procedures	4+

MIF continued on next page.

CTS REF	MANEUVER	FAM4502
23	Radio Communications	3+
24	Overhead/Break Entry	4+
25	Course Rules	1
26	Wave-off	4+
26	SSE Wave-off	4+
27	Landing Pattern	4+
27	No-Flap Pattern	4+
27	SSE Landing Pattern	4+
28	Landing	4+
28	NFL	4+
28	FFL	3+
28	SSE Landing	4+
29	Touch-and-Go	4+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+
	Special Syllabus Requirements	1

Chapter V

Instrument Training

Blk #	Media	Title	Events	Hrs	Blk Name
I01	Class	Instrument Brief/FMS	3	7.0	GPSFP

1. Prerequisites

   a. G0801-17 (Instrument Flight Rules – Phase I).

   b. G1090 (FRR Exam).

   c. FAM4390 (Fam Check Flight).

   d. I0101-3 (in order).

2. Events

I0101	Lect	GPS/Radio Instrument Procedures	3.0
I0102	MIL	ME GPS FMS	1.0
I0103	Lect/ FMS DTD	T-44C Flight Director Operation	3.0

3. Syllabus Notes. SNAs will complete the flight management systems Lab flight scenario during I0103.

4. Discuss Items. CRM callouts, runway/taxiway/parking signs and symbols, runway markings, local flying environment, pub bag, NOTAM, WX products, DoD FLIPs, FAR/AIM, Instrument Flight Manual, canned routes, coded departures, SIDs, Base OPS filing, FSS, communications, the six T's, GPS approach, RAIM, approach modes, local instrument approaches, and FMS desktop training device operation and scenario.

Blk #	Media	Title	Events	Hrs	H/X
I21	UTD	Basic Instruments	3	4.5	1.5

1. Prerequisites. FAM2103 (Cockpit Procedures).

2. Syllabus Notes. This block shall emphasize control and performance of basic instrument flight.

3. Special Syllabus Requirements. None.

4. Discuss Items

I2101
Instrument crosscheck, instrument takeoff, full-panel approach-to-stalls, stalls and spin recovery, high work maneuvers, and control and performance concept.

I2102
Unusual attitudes, pitot-static system, flight instrument characteristics, ESIS operation, wet compass characteristics, and departure procedures.

I2103
Anti-icing system, emergency or minimum fuel state, TERPs, IFR En route Supplement, Flight Information Handbook, NOTAMs, and autopilot/flight director usage.

5. Block MIF

CTS REF	MANEUVER	I2103
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
6	Headwork/Situational Awareness	3+
7	Basic Air Work	3+
8	Mission Planning/Briefing/ Debriefing	3+
10	Instrument Takeoff	3+
12	Departure	3+
14	Level Speed Change	4+

MIF continued on next page.

CTS REF	MANEUVER	I2103
15	Turn Pattern	4+
17	Approach to Stall	3+
21	In-Flight Planning/Clearance Compliance	3+
22	Cockpit Procedures	3+
23	Radio Communications	3+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	1
36	ESIS Approach to Stall	3+
36	ESIS Turns, Climbs/Descents	3+
36	Bravo Pattern	4+
36	Charlie Pattern	4+
36	Oscar Pattern	4+
36	ESIS Oscar Pattern	3+
36	Yankee Pattern (SSE)	3+
36	Unusual Attitude Recovery	4+
36	ESIS Unusual Attitudes	4+
47	Autopilot/Flight Director Operation	1

Blk #	Media	Title	Events	Hrs	H/X
I31	OFT	Instrument Approaches	4	6.0	1.5

1. Prerequisites

a. FAM3401 (Night Familiarization).

b. I0103 (Flight Director Operation).

2. Syllabus Notes

a. Students shall bring a completed DD-1801 for all RI events.

b. Autopilot shall not be used for approaches during the first two events in the block. Autopilot in the final two events shall be used and are limited to one fully coupled approach per event.

c. Simulator equipment-dependent emphasis items are:

(1) I3101. PAR, ASR, and ILS approaches.

(2) I3102. VOR, NDB, and TACAN approaches.

(3) I3103. LOC and LOC-BC approaches.

(4) I3104. RNAV (GPS) approaches (shall accomplish one of each) – HILO approach, "T" approach, and radar vectors. Shall see one LNAV approach, one LNAV/VNAV approach, and non-approach GPS holding.

d. Events shall have a minimum of three approaches per event and include at least two full procedure turn approaches.

e. Normal two-engine approaches should be emphasized in this block, but may introduce minor malfunctions (no ESIS or SSE).

f. Each event shall include a minimum of one approach with the flight director and one approach without the flight director.

g. Holding should be accomplished and graded on at least three different events.

h. All events shall include a missed approach; include at least two circling missed approaches in the block.

i. SNA in right seat shall be PM and graded accordingly, emphasizing CRM callouts and radio communications.

j. At least two non-precision approaches shall be conducted using the CDFA method.

3. Special Syllabus Requirements

I3102
CDFA Practice Approach.

4. Discuss Items

I3101
IAF and FAF procedures (6T's, descent, and lead turns), CRM callouts and techniques, GCA approach, PAR, ASR, and ILS configuration point, ASR recommended altitudes, NAVAID setup (takeoff, en route, approach, and missed approach), cockpit procedures (en route versus multiple terminal approaches, task management), and point-to-point navigation.

I3102
Holding, missed approach holding versus HILO procedures, VOR and TAC approach procedures, missed approach, circling procedures, circling missed approach (determining MAP and continuation beyond MAP), needle-only approach procedures, and continuous descent final approach (CDFA).

I3103
NATOPS Instrument Flight Manual, SOPs, local instructions, NATOPS, LOC and LOC-BC approaches (configuration, presentation, MA, and FAF Identification by other than DME), ESIS procedures, reverse sensing, and TERPS.

I3104
RNAV(GPS) approach types (LNAV, LNAV/VNAV, LPV, TAA, T's, and HILO), feeder fix selection of IAF, RNAV configuration point, RAIM, GPS sensitivity modes, autopilot and flight director, lost communications (FIH and LOA) , and SSE approach procedures.

5. Block MIF

CTS REF	MANEUVER	I3104
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
6	Headwork/Situational Awareness	3+
7	Basic Air Work	4+
8	Mission Planning/Briefing/ Debriefing	4+
10	Instrument Takeoff	4+
12	Departure	3+
21	In-Flight Planning/Clearance Compliance	3+
22	Cockpit Procedures	3+
23	Radio Communications	3+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	1
34	En route Procedures	3+
35	Point-to-Point	3+
38	Holding	3+
39	High Altitude Approach	3+
40	PAR	1
40	ILS	1
41,42	VOR	1
41,42	TAC	1
41,42	NDB	1
40-42	Needle Only Approach	1
42	Localizer	1
42	Localizer Back Course	1
42	RNAV/GPS Approach	1
42	ASR	1
42	CDFA Approach	3+

MIF continued on next page.

CTS REF	MANEUVER	I3104
39-43	Approach 1	3+
39-43	Approach 2	3+
39-43	Approach 3	3+
43	Circling Approach	2+
45	Circling Missed Approach	2+
47	Autopilot/Flight Director Operation	3+
48	FMS Operation	3+
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
I41	T-44C	Instrument Approaches	4	8.0	2.0

1. Prerequisites

a. FAM4502 (Night Familiarization).

b. I3104 (Instrument Approaches).

2. Syllabus Notes

a. IP shall make a note on the ATF if unable to accomplish emphasis items on appropriate flight. Weather, aircraft, and ATC equipment-dependent, emphasis items are:

(1) I4101. PAR, ASR, and ILS approaches.

(2) I4102. LOC and LOC-BC approaches.

(3) I4103. RNAV (GPS), VOR approaches.

(4) I4104. Discretion of the IP, but should include an SNA visual approach and an IP demonstration of SSE approach and wave-off.

b. Events should have a minimum of four approaches per event and include at least two procedure turn approaches. Normal two-engine approaches should be emphasized in this block, but may introduce minor malfunctions (no ESIS or SSE).

c. SNA shall see one LNAV and one LNAV/VNAV approach in the block.

d. Each event shall include a minimum of one approach with the flight director.

e. Holding should be accomplished and graded on at least two events, one of which should be GPS holding.

f. All events shall include a missed approach and should include at least two circling missed approaches in the block.

g. Minimum of one approach per event with IP as PF and SNA as PM, emphasizing CRM callouts and radio communications. To allow SNAs additional radio communication practice, one approach during each event should be flown with the SNA handling radio communications as PF.

h. SNAs shall bring a DD-1801 flight plan or FAA flight plan to correspond with the appropriate flight plan discuss item for each event in this block. This flight plan will be filled out as a practice flight plan to demonstrate the applicable knowledge of the different types of DD-1801 and FAA flight plans. Additionally, during each event brief, SNAs shall fill out a DD-1801 flight plan for the actual event profile as required.

i. At least two non-precision approaches shall be conducted using the CDFA method.

3. Special Syllabus Requirements

I4101
Coupled approach demo.

I4104
IP demonstrate SSE approach to SSE missed approach.

4. Discuss Items

I4101
Holding, GCA approach, PAR, ILS, and ASR configuration point, PAR versus ASR descent point, ASR recommended altitudes, NAVAID characteristics (ILS, LOC, LOC-BC (service volumes, operation principles, NATOPS procedures, and cockpit presentation)), ATC communication (clearance delivery, ground, tower, center, approach, tower, and ground), uncontrolled airfields (communications, canceling IFR, and procedures), holding in lieu of (HILO) PT approach, approach plate titles and notes, ATIS, AWOS, and ASOS, DD-175-1, and terminal area delay DD-1801.

I4102
Procedure turns (methods and types), instrument publications (DoD (NGA), FAA (NACO), Jeppesen, other), DPs (ODP, SID,vector, diverse, and uncontrolled field), VDP, CDFA, landing transition, low close-in obstacles, ILS, LOC, LOC-BC approach (configuration, presentation, MA, and FAF identification by other than DME), reverse sensing, and stopover DD-1801.

I4103
Point-to-point navigation; TERPS required obstacle clearance (ROC) for initial, intermediate, final, and circling phases; NAVAID characteristics (VOR, TACAN, and NDB (service volumes, operation principles, NATOPS procedures, and cockpit presentation)); VOR, TAC, and NDB approach procedures; missed approach (determining MAP and continuation beyond MAP), circling procedures and circling missed approach; procedure track (arc and radial combination and teardrop); VDP; and RNAV (FMS, GPS, DME/DME, VOR/DME, and INS), and FAA flight plan.

I4104
Pilot-controlled lighting; lost communications (FIH and LOA) (IFR – VMC versus IMC); wind shear; visual approach; contact approach; ESIS approach procedures; needle only approach procedures; SSE approach procedures; TERPS (FAA Order 8260.3B); Trouble T (OIS, low close-in obstacles, climb gradients, alternate routing, nonstandard takeoff weather minimums, and VCOA).

5. Block MIF

CTS REF	MANEUVER	I4104
1	General Knowledge/Procedures	4+
2	Emergency Procedures	3+
6	Headwork/Situational Awareness	3+
7	Basic Air Work	4+
8	Mission Planning/Briefing/ Debriefing	4+
9	Ground Operations	4+
10	Instrument Takeoff	4+
12	Departure	3+
21	In-Flight Planning/Clearance Compliance	3+
22	Cockpit Procedures	4+
23	Radio Communications	3+
28	Landing	4+
29	Touch-and-Go	4+
31	Pilot Flying/CRM	3+
32	Pilot Monitoring/CRM	3+
33	Clearing	4+
34	En route Procedures	3+
35	Point-to-Point	3+
37	En route Descent	1
38	Holding	3+
40	PAR	3+

MIF continued on next page.

CTS REF	MANEUVER	I4104
40	ILS	3+
41,42	VOR	3+
41,42	TAC	3+
41,42	NDB	1
40-42	Needle Only Approach	3+
42	Localizer	3+
42	Localizer Back Course	1
42	RNAV/GPS Approach	3+
42	ASR	3+
42	CDFA Approach	3+
43	Circling Approach	3+
44	Transition to Landing	3+
44	Visual Approach	1
45	Missed Approach	3+
45	Circling Missed Approach	3+
46	Radar Operation	1
47	Autopilot/Flight Director Operation	3+
48	FMS Operation	3+
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
I32	OFT	Emergency Instrument Approaches	4	6.0	1.5

1. Prerequisites. I4104 (Instrument Approaches).

2. Syllabus Notes

a. Students shall bring a completed DD-1801 for all RI events.

b. Autopilot shall not be used for approaches during the first two events in the block. Autopilot in the final two events shall be used and are limited to one fully-coupled approach per event.

c. Simulator equipment-dependent emphasis items are emergency procedures during:

(1) I3201. PAR, ASR, ILS approaches.

(2) I3202. LOC and LOC-BC approaches.

(3) I3203. VOR, NDB, and TACAN approaches.

(4) I3204. RNAV (GPS) approaches. Shall see one LNAV and one LNAV/VNAV approach.

d. Minimum of three approaches per event.

e. Each event shall include a minimum of one approach with the flight director.

f. Holding should be accomplished and graded on at least three different events, one of which shall be non-GPS.

g. All events shall include a missed approach; at least two circling missed approaches in the block.

h. SNA in right seat shall be PM and graded accordingly, emphasizing CRM callouts and radio communications.

i. At least two non-precision approaches shall be conducted using the CDFA method.

3. Special Syllabus Requirements. None.

4. Discuss Items

I3201
SSE approach procedures, SSE circling approach, procedure track (arc and radial combination and teardrop), emergency voice reports , departure procedures (SID, ODP, vector, diverse) and avionics system and malfunctions.

I3202
Weather filing criteria, turbulence and wind shear, icing, approach and landing minimums, ESIS approach, fuel system and malfunctions, in-flight damage and bird strikes.

I3203
En route weather filing facilities (FSS and PMSV), STARS (filing, planning, and lost comms), SSE missed approach, and electrical system and malfunctions.

I3204
SSE configuration point with or without VNAV, flight director malfunctions, autopilot malfunctions and disconnect procedures, landing gear system and malfunctions, and propeller system and malfunctions.

5. Block MIF

CTS REF	MANEUVER	I3204
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/ Debriefing	4+
10	Instrument Takeoff	4+
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
31	Pilot Flying/CRM	4+

MIF continued on next page.

CTS REF	MANEUVER	I3204
32	Pilot Monitoring/CRM	4+
34	En route Procedures	4+
35	Point-to-Point	3+
37	En route Descent	1
38	Holding	4+
40	PAR	1
40	ILS	1
40	SSE Precision Approach	3+
40-42	ESIS Approach	1
41,42	VOR	1
41,42	TAC	1
41,42	NDB	1
41,42	SSE Non-Precision Approach	3+
40-42	Needle Only Approach	4+
42	Localizer	1
42	Localizer Back Course	1
42	RNAV/GPS Approach	1
42	ASR	1
42	CDFA Approach	4+
39-43	Approach 1	4+
39-43	Approach 2	4+
39-43	Approach 3	4+
43	Circling Approach	3+
43	SSE Circling Approach	3+
44	Transition to Landing	1
45	Missed Approach	4+
45	SSE Missed Approach	3+
45	Circling Missed Approach	3+
47	Autopilot/Flight Director Operation	3+
48	FMS Operation	3+

Blk #	Media	Title	Events	Hrs	H/X
I42	T-44C	Emergency Instrument Approaches	4	8.0	2.0

1. Prerequisites. I3204 (Emergency Instrument Approaches).

2. Syllabus Notes

a. Each flight should consist of a mix of approaches flown in the I41 block.

b. Events should have a minimum of four approaches per event and include at least two procedure turn approaches. Emergency procedures should be emphasized in this block.

c. Each event shall include a minimum of one approach with the flight director. SNA shall see one LNAV and LNAV/VNAV approach in the block.

d. Holding should be accomplished and graded on at least two events, one of which should be GPS holding.

e. All events shall include a missed approach and should include at least two circling missed approaches in the block.

f. Minimum of one approach per event with IP as PF and SNA as PM, emphasizing CRM callouts and radio communications. To allow SNAs additional radio communication practice, one approach during each event should be flown with the SNA handling radio communications as PF.

g. During each event brief, SNAs shall fill out a DD-1801 flight plan for the actual event profile as required.

h. At least two non-precision approaches shall be conducted using the CDFA method.

3. Special Syllabus Requirements. None.

4. Discuss Items

I4201
SSE approach procedures (configuration, airspeeds and power in the descent, and descent rate), SSE circling approach and missed approach, emergency voice reports, propeller system and malfunctions, in-flight damage and bird strikes, ESIS approach, and simulated versus actual EP.

I4202
Needle only VOR and TAC approach procedures, landing minimums, flight director malfunctions, autopilot malfunctions and disconnect procedures, electrical system and malfunctions, anti-ice and deice system, windshield heating, and flight control system and malfunctions.

I4203
Procedure track (arc and radial combination and teardrop), VDP, CDFA, smoke and fume emergencies, smoke and fire of unknown origin, avionics system and malfunctions, and airfield diagrams and symbols.

I4204
Hazardous weather, pitot static system and malfunctions (alternate static and related avionics equipment), engine system and malfunctions (engine failure or fire after takeoff), STAR (filing, planning, and lost comms), fuel planning (required totals and reserve, and NATOPS performance information), fuel system and malfunctions, brake system and malfunctions, wake turbulence, wind shear, PULL UP warning, and weight and balance form F.

5. Block MIF

CTS REF	MANEUVER	I4204
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/ Debriefing	4+
9	Ground Operations	4+
10	Instrument Takeoff	4+
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
28	Landing	4+
29	Touch-and-Go	4+
31	Pilot Flying/CRM	4+

MIF continued on next page.

CTS REF	MANEUVER	I4204
32	Pilot Monitoring/CRM	4+
33	Clearing	4+
34	En route Procedures	4+
35	Point-to-Point	3+
37	En route Descent	1
38	Holding	4+
40	PAR	4+
40	ILS	4+
40	SSE Precision Approach	3+
40-42	ESIS Approach	3+
41,42	VOR	4+
41,42	TAC	4+
41,42	NDB	1
41,42	SSE Non-Precision Approach	3+
40-42	Needle Only Approach	4+
42	Localizer	4+
42	Localizer Back Course	1
42	RNAV/GPS Approach	4+
42	ASR	4+
42	CDFA Approach	4+
43	Circling Approach	4+
43	SSE Circling Approach	3+
44	Transition to Landing	4+
44	Visual Approach	1
45	Missed Approach	4+
45	SSE Missed Approach	3+
45	Circling Missed Approach	4+
46	Radar Operation	1
47	Autopilot/Flight Director Operation	3+
48	FMS Operation	4+

Blk #	Media	Title	Events	Hrs	H/X
I33	OFT	Mid-Stage Instrument Check	1	1.5	1.5

1. Prerequisites. I4204 (Emergency Instrument Approaches).

2. Syllabus Notes

a. This event will be an evaluation of IFR procedures, involving a representative cross section of maneuvers previously presented.

b. SNAs shall bring one DD-1801 flight plan per SNA. SNAs shall draft a flight plan that will execute the required maneuvers for the event.

c. Event shall have a minimum of three approaches and include at least one full procedure approach with exposure to all types of approaches.

d. Event shall include a minimum of one approach with the flight director.

e. Event shall be flown with military IP in right seat designated to give instrument check flights.

3. Special Syllabus Requirements. None.

4. Discuss Items. Operations limits quiz, IAF/FAF entry procedures (6 T's, descent, and lead turns), pressurization system and malfunctions, electrical system, and CNAF-M 3710.7 weather filing criteria.

5. Block MIF

CTS REF	MANEUVER	I3390
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	4+
9	Ground Operations	4+
10	Takeoff	1

MIF continued on next page.

CTS REF	MANEUVER	I3390
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
28	Landing	4+
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	4+
33	Clearing	4+
34	En route Procedures	4+
37	En route Descent	1
38	Holding	4+
39-43	Approach 1	4+
39-43	Approach 2	4+
39-43	Approach 3	4+
39-43	SSE Precision or Non-Precision Approach	3+
40-42	ESIS Approach	1
41/42	VOR	1
41/42	TAC	1
41/42	NDB	1
40-42	Needle Only Approach	1
42	Localizer	1
42	Localizer Back Course	1
42	RNAV/GPS Approach	1
42	ASR	1
43	Circling Approach	1
44	Transition to Landing	1
45	Missed Approach	1
45	SSE Missed Approach	3+
47	Autopilot/Flight Director Operation	3+
48	FMS Operation	4+
49	Terminal Procedures	3+

Blk #	Media	Title	Events	Hrs	H/X
I43	T-44C	Cross-Country	4	16.0	4.0

1. Prerequisites

a. I3390 (Mid-stage Instrument Check).

b. G0990 (IFR Final Exam).

c. G1090 (FRR Exam).

2. Syllabus Notes

a. Shall be conducted out of the local area concentrating on en route cross-country navigation with a minimum of two approaches conducted per event or a total of eight approaches. Flight director shall be used for one approach per event. Autopilot with flight director should be used during this block.

b. Numbered approaches shall expose the student to all types of approaches. A visual approach shall be flown at least once during the block.

c. Should be flown above FL180 to emphasize high altitude en route navigation.

d. The CNATRAINST 1500.4L limit for the ***maximum SNA instructional flight hours in one day*** is removed for this block only. SNA shall not exceed the 12-hour workday.

e. I4301 and I4303 are right seat events for SNA.

f. Should fly a SID and a STAR during the block.

g. Ownship should be utilized on ForeFlight during cross-country operations.

3. Special Syllabus Requirements

I4301
Demonstrate weather radar procedures.

I4302
FMS Assisted Approach demo.

4. Discuss Items

I4301
CRM Seven Skills, callouts, ATC communications (clearance delivery, ground, tower, center, approach, tower, and ground), pre-flight planning (nav bag, fuel planning and packet, weight and balance, NOTAMS (destination, alternates, and en route), CNAF-M 3710.7 weather filing criteria, DD-1801, DD-175-1, OPARS, AIRMETs, and SIGMETs), unfamiliar field operations, and MFD operation.

I4302
DPs (ODP, SID, vector, diverse), checklist management, en route charts, airfield markings and lighting, STARs (filing, planning, and lost comms), en route weather facilities (FSS and Metro), airport diagrams and symbols, circling and circling missed approach (TAS considerations), securing and RON procedures.

I4303
FAA flight plan, contacting FSS, performance data, anti-icing systems, RNP, lost communications (en route, VMC versus IMC), and course deviations for weather.

I4304
Filing in-flight, SSE en route, FMS en route navigation, wake turbulence, arrival transition, and procedure turn (holding technique).

5. Block MIF

CTS REF	MANEUVER	I4304
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	4+
9	Ground Operations	4+
10	Takeoff	4+
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+

MIF continued on next page.

CTS REF	MANEUVER	I4304
28	Landing	4+
29	Touch-and-Go	1
30	SSE Full-Stop	1
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	4+
33	Clearing	4+
34	En route Procedures	4+
37	En route Descent	4+
39-43	Approach 1	4+
39-43	Approach 2	4+
44	Transition to Landing	4+
44	Visual Approach	3+
45	Missed Approach	1
45	SSE Missed Approach	1
45	Circling Missed Approach	1
46	Radar Operation	4+
47	Autopilot/Flight Director Operation	4+
48	FMS Operation	4+
49	Terminal Procedures	4+
	Special Syllabus Requirements	1

Blk #	Media	Title	Events	Hrs	H/X
I34	OFT	Advanced Emergency Instrument Approaches	2	3.0	1.5

1. Prerequisites. I4304 (Cross Country).

2. Syllabus Notes

a. This event is a scenario-based, comprehensive review of instrument procedures, involving a representative cross section of IFR approaches, and will prepare the student for initial instrument rating. It should include degradation of aircraft systems, avionics automation, and IP, PM CRM. It must emphasize the following: CRM, PIC decision making, situational awareness, instrument proficiency, and multi-engine airmanship.

b. To allow SNAs additional radio communication practice, one approach during event should be flown with the SNA handling radio communications as PF.

c. The event should have a minimum of three approaches per event, including a minimum of two non-local. Numbered approaches shall expose the student to all types of approaches.

d. At a minimum, the following should be flown in the block: one approach with glideslope/glide-path guidance, one VOR or TAC or NDB approach, one approach with use of a flight director, and one approach without the use of a flight director.

e. CI will instruct event.

3. Special Syllabus Requirements. None.

4. Discuss Items

I3401

V_{mca}, V_{mcg}, PIC decision making, WW versus convective SIGMET, weather considerations during emergencies, approach lighting, weight and balance form F, and other discussion items at the discretion of the IP.

I3402

Memory items and deferred emergencies, NATOPS Procedures and situations not covered by NATOPS, changing flight plan in-flight, ATC priority handling, adaptability/flexibility during emergencies, and other discussion items at the discretion of the IP.

5. Block MIF

CTS REF	MANEUVER	I3402
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	4+
9	Ground Operations	1
10	Takeoff	4+
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
28	Landing	4+
30	SSE Full-Stop	1
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	4+
33	Clearing	4+
34	En route Procedures	4+
35	Point-to-Point	4+
39-43	Approach 1	4+
39-43	Approach 2	4+
39-43	Approach 3	4+
40	SSE Precision Approach	3+
40-42	ESIS Approach	3+
41/42	SSE Non-Precision Approach	3+
45	Missed Approach	4+
45	SSE Missed Approach	4+
45	Circling Missed Approach	1

MIF continued on next page.

CTS REF	MANEUVER	I3402
46	Radar Operation	1
47	Autopilot/Flight Director Operation	4+
48	FMS Operation	4+
49	Terminal Procedures	4+

Blk #	Media	Title	Events	Hrs	H/X
I35	OFT	Instrument Emergency Procedures/ CRM	2	3.0	1.5

1. Prerequisites. I3204 (Emergency Instrument Approaches).

2. Syllabus Notes

a. I3501 will include various aircraft and instrument malfunctions. CI will instruct event.

b. I3502 will consist of a real world scenario commencing at crew brief or take-off. Emphasis will be placed on procedural knowledge, judgment, and crew resource management skills. PM SNA shall be graded on CRM, radio communications, and callouts. The scenario will be carried through to a logical conclusion. IP or CIS, CRM-I or CRM-F, shall evaluate SNA as per the OPNAVINST 1542.7E.

c. Upon successful completion I3502, CRM-I or CRM-F shall record flight evaluation on CRM Training and Evaluation Record for inclusion in SNA's NATOPS jacket. IP or CIS shall include on ATF "Successful completion of initial CRM evaluation as per the COMNAVAIRFORINST 1542.7D."

3. Special Syllabus Requirements. None.

4. Discuss Items

I3501
Simulated scenario background.

I3502
CRM and simulated scenario background.

5. Block MIF

CTS REF	MANEUVER	I3502
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+

MIF continued on next page.

CTS REF	MANEUVER	I3502
8	Mission Planning/Briefing/ Debriefing	4+
10	Instrument Takeoff	4+
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	4+
39-43	Approach 1	4+
39-43	Approach 2	4+
39-43	Approach 3	4+
39-43	SSE Precision or Non-Precision Approach	4+
47	Autopilot/Flight Director Operation	1

Blk #	Media	Title	Events	Hrs	H/X
I44	T-44C	Advanced Emergency Instrument Approaches	3	6.0	2.0

1. Prerequisites

a. I3402 (Advanced Emergency Instrument Approaches).

b. I3502 (Emergency Instrument Approaches).

2. Syllabus Notes

a. This block is a scenario-based, comprehensive review of instrument procedures, involving a representative cross section of IFR approaches, and will prepare the student for initial instrument rating. Each event should include degradation of aircraft systems, avionics automation, IP and PM CRM. These events must emphasize the following: CRM, PIC decision making, situational awareness, instrument proficiency, and multi-engine airmanship.

b. SNAs shall contact the IP prior to the event for an assignment related to flight. If the IP is unavailable, contact the CDO for an assignment. SNAs shall bring one DD-1801 flight plan per SNA and one DD-175-1 per aircraft for their planned profile to every brief. SNAs shall draft a flight plan that will execute the required maneuvers for the event(s).

c. Minimum of one approach per event with IP as PF and SNA as PM, emphasizing CRM callouts and radio communications. To allow SNAs additional radio communication practice, one approach during each event should be flown with the SNA handling radio communications as PF.

d. Should have a minimum of four approaches per event, including a minimum of two non-local. Numbered approaches may include ILS, LDA, LOC, LOC BC, RNAV, VOR, TAC, NDB, Visual, PAR, ASR, Needle-only, ESIS only, HI approaches, and circling.

e. At a minimum, the following should be flown in the block: one approach with glideslope/glide-path guidance, one VOR or TAC or NDB approach, one approach without the use of a flight director.

f. Any approach that does not meet MIF SHALL be remediated by flying that specific type of approach prior to end of block.

3. Special Syllabus Requirements. None.

4. Discuss Items

I4401
Airspace and RVSM, weight and balance form F, "land as soon as practicable versus possible," RNP, and other discussion items at the discretion of the IP.

I4402
Student brief CRM case study, threat and error management, CRM, ORM, and other discussion items at the discretion of the IP.

I4403
Density altitude considerations (true airspeed, descents required, and circling rate of turn), instrument rating request form CNAF-M 3710.7 and other discussion items at the discretion of the IP.

5. Block MIF

CTS REF	MANEUVER	I4403
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	4+
9	Ground Operations	4+
10	Takeoff	4+
12	Departure	4+
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
28	Landing	4+
29	Touch-and-Go	4+
30	SSE Full-Stop	4+
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	4+

MIF continued on next page.

CTS REF	MANEUVER	I4403
33	Clearing	4+
34	En route Procedures	4+
35	Point-to-Point	4+
38	Holding	4+
39-43	Approach 1	4+
39-43	Approach 2	4+
39-43	Approach 3	4+
40	SSE Precision Approach	4+
40-42	ESIS Approach	4+
40-42	Needle Only Approach	4+
41/42	SSE Non-Precision Approach	4+
43	Circling Approach	4+
43	SSE Circling Approach	4+
44	Transition to Landing	4+
44	Visual Approach	4+
45	Missed Approach	4+
45	SSE Missed Approach	4+
45	Circling Missed Approach	4+
47	Autopilot/Flight Director Operation	4+
48	FMS Operation	4+

Blk #	Media	Title	Events	Hrs	H/X
I45	T-44C	NATOPS Instrument Check Flight	1	2.0	2.0

1. Prerequisites. I4403 (Advanced Emergency Instrument Approaches).

2. Syllabus Notes

a. This event will be a comprehensive evaluation of IFR procedures, involving a representative cross section of instrument approaches and emergency procedures.

b. Complete a minimum of four approaches to include one precision and one non-precision approach. Numbered approaches may include ILS, LDA, LOC, LOC BC, RNAV, VOR, TAC, NDB, Visual, PAR/ASR, Needle-only, ESIS only, HI approaches, and circling. One approach shall be flown by the IP with the student acting as PM.

c. SNAs are required to bring a completed instrument rating request form to the brief.

d. IFR final must be within previous 60 days.

e. Event shall be flown per the CNAF-M 3710.7 and the NATOPS Instrument Flight Manual. Successful completion shall lead to issuance of CNAF-M 3710.7 initial Standard Instrument Rating.

3. Special Syllabus Requirements. None.

4. Discuss Items. Instrument rating request form CNAF-M 3710.7, Trouble T (OIS, low close-in obstacles, climb gradients, alternate routing, nonstandard takeoff weather minimums, and VCOA), operations limits quiz, and additional discussion items at the discretion of the IP.

5. Block MIF

CTS REF	MANEUVER	I4590
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/Debriefing	4+

MIF continued on next page.

CTS REF	MANEUVER	I4590
9	Ground Operations	4+
10	Takeoff	1
12	Departure	4+
34	En route Procedures	4+
21	In-Flight Panning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
28	Landing	4+
29	Touch-and-Go	1
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	4+
33	Clearing	4+
36	3710/2 Instrument Maneuvers	4+
37	En route Descent	1
38	Holding	1
39	High Altitude Approach	1
39-43	Approach 1	4+
39-43	Approach 2	4+
39-43	Approach 3	4+
39-43	SSE Precision or Non-Precision Approach	4+
40	SSE Precision Approach	1
41/42	SSE Non-Precision Approach	1
40-42/ 44	ESIS Approach	4+
44	Transition to Landing	4+
44	Visual Approach	1
45	Missed Approach	1
45	SSE Missed Approach	4+
45	Circling Missed Approach	1
47	Autopilot/Flight Director Operation	4+
48	FMS Operation	4+
49	Terminal Procedures	4+

Chapter VI

Navigation Training

This chapter does not apply to Intermediate E-2 MPTS.

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Chapter VII

Formation Training

Blk #	Media	Title	Events	Hrs	Blk Name
F01	Lect	Maritime Formation Procedures	1	1.0	FORMFP

1. Prerequisites. I4590 (NATOPS Instrument Check Flight).

2. Events

F0101	Lect	Maritime Formation Procedures	1.0

3. Syllabus Notes. None.

4. Discuss Items. None.

Blk #	Media	Title	Events	Hrs	H/X
F41	T-44C	Maritime Formation Fundamentals	1	2.0	2.0

1. Prerequisites. F0101 (Maritime Formation Procedures).

2. Syllabus Notes

   a. Complete formation sequence as per the FTI and formation brief guide.

   b. RV Delta (Point Parallel) Rendezvous should be conducted where practicable.

3. Special Syllabus Requirements. None.

4. Discuss Items. Formation flight procedures, flight sequence, and formation emergency procedures.

5. Block MIF

CTS REF	MANEUVER	F4101
1	General Knowledge/Procedures	4+
2	Emergency Procedures	4+
6	Headwork/Situational Awareness	4+
7	Basic Air Work	4+
8	Mission Planning/Briefing/ Debriefing	4+
9	Ground Operations	4+
10	Formation Takeoff	1
12	Departure	1
21	In-Flight Planning/Clearance Compliance	4+
22	Cockpit Procedures	4+
23	Radio Communications	4+
24	Traffic Entry	1
25	Course Rules	1

MIF continued on next page.

CTS REF	MANEUVER	F4101
28	Landing	1
31	Pilot Flying/CRM	4+
32	Pilot Monitoring/CRM	1
33	Clearing	4+
48	FMS Operation	1
50	Running Rendezvous	3+
50	Parade Position	3+
50	Parade Turns	3+
50	Cross-under	3+
50	Free Cruise	3+
50	Breakup and Rendezvous	3+
50	Maritime Lead Change	3+
50	Maritime Lead	3+
51	Alternate Rendezvous	1

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Chapter VIII

Tactical Training

This chapter does not apply to Intermediate E-2 MPTS.

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Chapter IX

Course Training Standards (CTS)

1. Purpose. These standards outline the tasks and proficiency required of graduates of this syllabus.

2. Student Duties and Responsibilities

a. Plan the mission.

b. Ensure the aircraft is preflighted, inspected, and equipped for the assigned mission.

c. Operate the aircraft to accomplish the mission using sound judgment and airmanship.

3. General Proficiency Standards

a. Achieve training standards for VMC maneuvers in conjunction with visual clearing.

b. Unless otherwise specified, use ***Basic Air Work (BAW)*** standards for all items with altitude, airspeed, or heading parameters.

c. "Standard" equates to ***good*** (G/4).

d. Aircraft control must be smooth and positive. Performance may be within CTS and still not warrant a grade of ***good*** if control inputs are delayed, erratic, imprecise, or inappropriate. Slight deviations in establishing or maintaining the proper or desired aircraft attitude or position may occur during the maneuver being performed.

e. Momentary deviations outside CTS that do not compromise flight safety are acceptable if subsequent corrections are timely.

f. Procedural knowledge and application must comply with applicable directives and allow efficient mission accomplishment. If individual tasks require pre-mission planning, the standards from ***Mission Planning*** apply.

4. Execution. The MIF regulates SNA progression to meet required standards prior to phase completion. Instructor Pilots shall evaluate SNA performance against these standards.

5. Job Tasks. Specific performance and standards required are described as follows:

BEHAVIOR STATEMENT	STANDARDS
GRADED ITEM	
● A brief description of the behavior, required action, and/or conditions.	● The specific standards for the action. May be read as "The SNA …"

6. Graded Items. The MIF for specific graded items varies for each stage. Several items are graded on all complete syllabus events. The standards for these universally graded items are listed first.

7. Course Training Standards

BEHAVIOR STATEMENT	STANDARDS
1. General Knowledge/Procedures	
● Maintain working knowledge of all appropriate flight training instructions and directives.	● Recites, discusses, or performs all applicable items essential to the operation of the airplane.
2. Emergency Procedures	
● Maintain in-depth knowledge of NATOPS and appropriate directives.	● Correctly analyzes situation. ● Performs and recites critical action steps from memory. ● Uses checklist when conditions permit. ● Completes procedures in a timely manner.
3. Start Malfunctions	
● Maintain in-depth knowledge of NATOPS and appropriate directives.	● Correctly analyzes situation. ● Performs and recites critical action steps from memory. ● Uses checklist when conditions permit. ● Completes procedures in a timely manner.
4. Windmilling Air-Start	
● Comply with the FTI and NATOPS while maintaining situational awareness sufficient for flight safety.	● Understands instructions, demonstrations, and explanations. ● Foresees and avoids possible difficulties. ● Remains alert and spatially oriented.
5. Starter-Assisted Air-Start	
● Comply with the FTI and NATOPS while maintaining situational awareness sufficient for flight safety.	● Understands instructions, demonstrations, and explanations. ● Foresees and avoids possible difficulties. ● Remains alert and spatially oriented.

BEHAVIOR STATEMENT	STANDARDS
6. Headwork/Situational Awareness	
● Comply with the FTI and NATOPS while maintaining situational awareness sufficient for flight safety.	● Understands instructions, demonstrations, and explanations. ● Foresees and avoids possible difficulties. ● Remains alert and spatially oriented.
7. Basic Air Work	
● Establish and maintain desired altitude, airspeed, and heading during flight.	● Maintains aircraft within +/-100 feet, 10 KIAS, 10° of heading. ● Appropriately uses power, attitude, and trim. ● Levels off within 100 feet of desired altitude. ● Accomplishes within ±10 seconds of correct time as applicable.
8. Mission Planning/Briefing/Debriefing	
● Flight room or base operations environment.	● Plans mission in a timely manner to meet all maneuver requirements. ● Acquires appropriate flight planning and weather data. ● Clearly defines the mission overview and mission goals with NATOPS briefing guide. ● Effectively uses the mission debriefing to reinforce skills and identify key points in mission performance.
9. Ground Operations	
● Begins when departing for the aircraft and ends when the power is applied for takeoff. ● Begins again when aircraft clears the runway and continues until power is advanced for a subsequent takeoff or when post-flight duties are complete and the aircrew is clear of the aircraft.	● Complies with NATOPS and training directives. ● Determines aircraft status and TOLD. ● Properly operates aircraft systems on ground. ● Ensures clearance of line personnel, ground equipment, and other aircraft using appropriate signals. ● Taxies aircraft at speeds commensurate with traffic and surface conditions. ● Maintains taxiway boundaries (including hold-short) and gives way to other aircraft as appropriate.

BEHAVIOR STATEMENT	STANDARDS
10. Takeoff	
• Begins when advancing power for takeoff and ends when aircraft is safely airborne, gear and flaps are retracted, and climb power and airspeed are established.	• Checks aircraft performance per NATOPS. • Maintains ±10 feet of centerline. • Rotates at VR +5/-0 knots. • Transitions to instrument flight, if required. • Smoothly accelerates to appropriate climb speed. • Maintains runway situational awareness to include Go/No-Go criteria. • Maintains appropriate separation from lead, if applicable.
11. Aborted Takeoff	
• Begins from the recognition of malfunction or possible abort condition. • Ends when aircraft is stopped and radio call is completed.	• Handles per NATOPS. • Verbalizes "abort" internally and on the radio. • Remains ±10 feet of centerline. • Brings aircraft smoothly to a full-stop utilizing brakes and reverse.
12. Departure	
• Begins when climb airspeed is established and ends when published departure is complete or established in assigned working area. • If no published departure, begins when initiation pitch change for level-off.	• Complies with ATC, departure, flight plan clearance or course rules, as appropriate.
13. Dynamic Engine Cut	
• Begins with initial pitch and power setting. • Ends when aircraft is climbing trimmed at Vyse (minimum Vxse), on takeoff heading, and Emergency Shutdown Checklist has been executed.	• Executes maneuver as per the NATOPS and FTI. • Maintains ±10 degrees of heading; demonstrates appropriate rudder use. • Does not descend below initial altitude, and airspeed is never below 91 knots.
14. Level Speed Change	
• Perform level speed change per the FTI.	• Perform within Basic Air Work CTS.

BEHAVIOR STATEMENT	STANDARDS
15. Turn Pattern	
● Perform turn pattern per the FTI.	● Maintains:±100 feet, ±5° bank, ±10 knots, rolls out ±5°.
16. Slow Flight	
● Perform slow flight per the FTI.	● Airspeed +5,-0 KIAS. ● Maintains BAW.
17. Approach to Stall	
● Begins with initial pitch and power setting. ● Ends when aircraft is stabilized in straight- and-level flight, on climb profile (Vx or Vy), or in position for the next maneuver.	● Minimizes altitude loss. ● Once approach-to-stall is stopped, avoids secondary approach-to-stall indications. ● Performs smooth, controlled recovery.
18. SSE Wave-off at Altitude	
● Begins with initial pitch and power setting. ● Ends when aircraft is established in a clean climb, minimum of Vxse (preferably Vyse) with aircraft trimmed and in balanced flight.	● Executes maneuver as per the NATOPS and FTI Procedures.
19. Emergency Descent	
● Performs procedure as per the NATOPS and FTI. ● Begins with initial power reduction. ● Ends when straight and level, and recovery is complete.	● Flies maneuver at maximum rate of descent for configuration. ● Initiates and completes recovery per FTI at prescribed altitudes.
20. Power-on Ditch/SSE Ditch/ Power-off Ditch	
● Executes maneuver as per the NATOPS and FTI. ● Begins when the decision to ditch is made. ● Ends when the aircraft arrives at simulated impact.	● Performs simulated impact at wings-level within ±20 degrees of ditch heading and +5/-0 knots. ● Maintains rate of descent at or below: ► 100 feet per minute, power-on. ► 100 feet per minute, SSE. ► 500 feet per minute, power-off.

BEHAVIOR STATEMENT	STANDARDS
21. In-Flight Planning/Clearance Compliance	
• Demonstrates general understanding of mission flow and area orientation, both vertically and horizontally, recognizing and avoiding potential hazards.	• Demonstrates effective time management. • Accomplishes mission maneuver items. • Complies with ATC clearance. • Remains within assigned airspace.
22. Cockpit Procedures	
• Prioritizes and manages crew tasks during mission profile; ensures complete checklist discipline and the following of all standard operating procedures.	• Correctly prioritizes multiple tasks; uses all available resources to manage workload. • Accomplishes all required normal and emergency checklists for each phase of flight; completes checklists in a timely manner with all items addressed. • Completes all ground checklists and is prepared for takeoff in a timely manner.
23. Radio Communications	
• Performs verbal communications during mission profile.	• Uses precise, properly formatted radio calls with standard terminology. • Acknowledges all communications. • Understands and prioritizes transmissions in a multiple communications environment. • Asks for and provides clarification when necessary.
24. Overhead/Break Entry	
• Begins when commencing the break. • Ends when arriving wings-level on the downwind leg at pattern altitude.	• Complies with FTI procedures. • Maintains altitude +100/-50 feet of FTI requirements.
25. Course Rules	
• Begins from VFR takeoff. • Ends when commencing the break, established on straight-in final, or wings-level on downwind.	• Accomplishes as per the NATOPS, FTI, and Course Rules, as appropriate.

BEHAVIOR STATEMENT	STANDARDS
26. Wave-off/SSE Wave-off	
● Begins when advancing power levers. ● Ends when aircraft is safely airborne, gear and flaps are retracted, appropriate airspeed is established, and crosswind turn is initiated.	● Complies with NATOPS and FTI procedures. ● Initiates wave-off when: ► Aircraft requires more than 30-degree AOB to avoid overshooting final. ► Landing clearance not received by short final. ► Directed. ► Unsafe gear indication after rolling onto final. ► Aircraft is not in a safe position to make a safe landing.
27. Landing Pattern/No-Flap Pattern/SSE Landing Pattern	
● Begins when commencing the break, initiating crosswind, wings-level on downwind leg, or established on straight-in final. ● Ends at landing phase or when adding power for wave-off.	● Complies with NATOPS and FTI procedures. ● Maintains +5/-0 knots of FTI established pattern airspeeds prior to threshold. ● Dual engine, arrives at threshold +5/-0 knots of VREF speed. ● SSE, Maintains +5/-0 knots of FTI-established pattern airspeeds prior to crossing threshold. ● Altitude ±50 feet of FTI requirements throughout pattern.
28. Landing/NFL/FFL/SSE Landing/Right Seat Landing	
● Begins when crossing the threshold or initiating the round-out, whichever occurs first. ● Ends when slowed to a safe taxi speed or when advancing power for touch-and-go takeoff.	● Dual engine, arrives at threshold +5/-0 knots of VREF speed. ● SSE, arrives at threshold +10/0 knots of VREF speed. ● Touches down in the prescribed landing zone. ● Lands and maintains within ±10 feet of runway centerline. ● Touches down <300 feet/min. ● Applies proper crosswind correction.
29. Touch-and-Go	
● Begins when advancing power after landing. ● Ends when aircraft is safely airborne, gear and flaps are retracted (if appropriate), and appropriate airspeed is established.	● Accomplishes as per the NATOPS and FTI. ● Maintains runway centerline ±10 feet.

BEHAVIOR STATEMENT	STANDARDS
30. SSE Full-Stop	
• Begins at touchdown on the runway and ends when the aircraft has come to a slow taxi speed on the runway.	• Performs maneuver per the FTI. • Maintains centerline ±25 feet.
31. Pilot Flying/Crew Resource Management (CRM)	
• Decision Making. • Assertiveness. • Mission Analysis. • Communications (PM will normally make all external radio communications, graded item 19). • Leadership. • Adaptability and Flexibility. • Situational Awareness (individually graded item). • As PF, embraces Threat and Error Management and callouts, acting as a complete team member.	• Gathers available data before arriving at final decision; clearly states decisions to the crew; and provides rationale for decisions. • Displays assertive behavior when necessary and accepts assertive behavior from other crewmembers. • Assesses risks and makes decisions; identifies probable contingencies and alternatives. • Facilitates effective, open, and clear internal communications; uses exact checklist and callout verbiage as per the FTI and NATOPS. • Recognizes and eliminates hazardous attitudes in self and other crewmembers; resolves conflict in a positive manner. • Provides positive leadership to the crew; encourages crew participation in the decision-making process. • Adapts to meet new situational demands. • Demonstrates the ability to maintain awareness of what is happening on the ground, in the air, and with other crewmembers; copes with any subsequent mission impact as a result of these happenings.

BEHAVIOR STATEMENT	STANDARDS
32. Pilot Monitoring/CRM	
● Decision Making. ● Assertiveness. ● Mission Analysis. ● Communications (PM will normally make all external radio communications, graded item 19). ● Leadership. ● Adaptability and Flexibility. ● Situational Awareness (individually graded item). ● As PM, embraces Threat and Error Management and callouts, acting as a complete team member.	● Gathers available data before arriving at final decision; clearly states decisions to the crew; and provides rationale for decisions. ● Displays assertive behavior when necessary and accepts assertive behavior from other crewmembers. ● Assesses risks and makes decisions; identifies probable contingencies and alternatives. ● Facilitates effective, open, and clear internal communications; uses exact checklist and callout verbiage as per the FTI and NATOPS. ● Recognizes and eliminates hazardous attitudes in self and other crewmembers; resolves conflict in a positive manner. ● Provides positive leadership to the crew; encourages crew participation in the decision-making process. ● Adapts to meet new situational demands. ● Demonstrates the ability to maintain awareness of what is happening on the ground, in the air, and with other crewmembers; copes with any subsequent mission impact as a result of these happenings.
33. Clearing	
● Begins at engine start and ends with both engines shut down and parking brake set.	● Accomplishes flight deck and mission tasks while remaining visually and aurally alert to and avoiding other in-flight and ground obstacles. ● Effectively uses accepted visual clearing techniques to avoid conflicts. ● Effectively uses radios and other crewmembers to aid in clearing.

BEHAVIOR STATEMENT	STANDARDS
34. En route Procedures	
• Begins when established at assigned altitude, airspeed, and power setting. • Ends with initial power reduction for descent or entering en route holding.	• Updates and validates planned time and fuel computations as required to safely and efficiently accomplish the mission. • Effectively uses FSS, PMSV, and ATIS as required. • Maintains course centerline between all NAVAIDs and fixes with minor deviations (if IFR).
35. Point-to-Point	
• Conduct a point-to-point per the FTI.	• Executes point-to-point within ±2 NM.
36. Basic Instrument Maneuver	
• Begins with initial pitch and power setting. • Ends when aircraft is stabilized in straight- and-level flight or in position for the next maneuver.	• Accomplishes all maneuvers per the FTI. • Maintains Basic Aircraft Control parameters. • Maintains VSI within ±200 FPM of established parameters, if applicable.
37. En route Descent	
• Begins with initial power reduction at cruise. • Ends when crossing the holding fix, IAF, or established on radar vectors cleared for approach.	• Complies with ATC, STAR, flight plan clearance. • Arrives at assigned or briefed altitude with sufficient time to slow and configure (if required) prior to the terminal fix.
38. Holding	
• Begins when crossing the holding fix and ends when departing the holding pattern for a subsequent fix or the approach.	• Computes proper entry turn. • Estimates wind direction and applies appropriate corrections. • Establishes and maintains aircraft with holding airspace.
39. High Altitude Approach (Penetration)	
• Begins when crossing the high IAF and ends at the FAF.	• Complies with all altitude restrictions. • Maintains airspeed as per the FLIP, NATOPS, and FTI.

BEHAVIOR STATEMENT	STANDARDS
40. Precision Approach	
● Begins when cleared for the approach on radar vectors or when intercepting glide-path on a published approach procedure. ● Ends at transition to landing or applying power to execute a missed approach and wave-off.	● ILS approach: ►Maintains ±1 dot width of localizer and glideslope. ● PAR approach: Does not exceed "well above or below glide-path" or "well left or right of course" and complies with the controller's instructions in a timely manner. ● Maintains +5/-0 knots of approach airspeed. ● Maintains arcs ±.5 DME.
41. Non-Precision Procedure Turn/Arc	
● Begins on crossing IAF on a published approach procedure. ● Ends when crossing FAF on a published approach procedure.	● Executes course reversal as per the NATOPS, FTI, and FAR/AIM, as appropriate. ● Adjusts outbound leg to stay inside "remain within distance." ● Maintains +10/-0 knots of approach airspeed. ● Maintains arcs ±.5 DME.
42. Non-Precision Approach	
● Begins when cleared for the approach on radar vectors or when crossing the FAF on a published approach procedure. ● Ends at transition to landing or applying power to execute a missed approach and wave-off.	● FAF to MAP: ►Begins timing within ±5 seconds if appropriate. ► +10/-0 knots of approach airspeed. ►Course ±1 dot width. ● Arrives at the MDA prior to MAP in a safe position to make a normal visual descent to land. ● MDA +100/-0 feet. ● NDB final approach: Maintains ±5° bearing. ● ASR approach: Does not exceed "well left or right of course" and complies with the controller's instructions in a timely manner.
43. Circling Approach/SSE Circling Approach	
● Begins when initiating the circle and ends at landing phase or applying power to execute a missed approach and wave-off.	● Accomplishes circle as per the FTI and FAR/AIM. ● Maintains circling altitude +100/-0 feet. ● Maintains circling airspeed +10/-0 knots.

BEHAVIOR STATEMENT	STANDARDS
44. Transition to Landing/Visual Approach	
● Begins when departing the MDA or DH on a visual glide-path to the runway and ends at landing phase.	● Maintains a normal visual glide-path to the runway. ● Follows visual approach guidance as appropriate, i.e., VASI, PAPI, etc. ● Arrives at threshold +5/-0 knots of VREF speed.
45. Missed Approach/SSE Missed Approach/Circling Missed Approach	
● Begins when advancing power and ends when aircraft is safely climbing, gear and flaps are retracted, appropriate airspeed is established, and missed approach or climb-out instructions are complied with or crosswind turn is initiated.	● Accomplishes as per the FTI and NATOPS. ● Complies with FLIP missed approach procedures or climb-out instructions, as appropriate. ● Establishes runway or assigned heading ±5°, if appropriate. ● Initiates when field not in sight and ► Non-precision ▪ Inside FAF and full scale CDI deflection. ▪ At specified MAP DME. ▪ At expiration of timing in the absence of DME. ► Precision, first of ▪ Decision height, ▪ Controller-directed,
46. Radar Operation	
● Understands and applies system operation and limitations.	● Demonstrates ability to use radar for weather observation and avoidance.
47. Autopilot/Flight Director Operation	
● Understands and applies system operation and limitations.	● Correctly and appropriately uses autopilot/FD in the horizontal and vertical modes to improve pilot task loading or clearing. ● With autopilot/FD engaged, maintains aircraft control within course training standards for the given phase of flight.

BEHAVIOR STATEMENT	STANDARDS
48. FMS Operation	
• Understands and applies system operation and limitations.	• Effectively and accurately programs and navigates using the FMS. • Able to store and retrieve flight plans. • Able to use system features to enhance situational awareness. • Accomplishes tasks in a timely manner.
49. Terminal Procedures	
• Encompasses procedures in the Terminal Area such as Point-to-Point, Holding, Arcs, etc. that can be separated from the approach itself.	• Adheres to the CTS standards for those items as delineated above.
50. Maritime Formation Maneuvers	
• Begins with formation taxi and ends when the formation is split up for recovery or in the pattern.	• Accomplishes maneuvers as per the mission brief and the FTI. • Demonstrates a working knowledge of formation procedures as established in the FTI. • Maintains wingman position stabilized with safe separation between aircraft. • Demonstrates wingman consideration while lead.
51. USN Aerial Refueling Maneuvers (AR)	
• Begins when the aircraft is maneuvered to the Aerial Refueling Initial Point or the Aerial Refueling Control Point and ends after departing the aerial refueling track.	• Accomplishes maneuvers as per the mission brief and the FTI. • Demonstrates a working knowledge of aerial refueling procedures as established in the FTI. • Maintains a stabilized rendezvous with safe separation between aircraft. • Demonstrates stabilized pre-contact position while receiver. • Maintains precise control of the aircraft while demonstrating aerial refueling limits.

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Chapter X

Master Materials List

1. Individually Issued Materials

NOMENCLATURE	IDENTIFICATION	QTY PER STUDENT
a. T-44C E-2 Intermediate Multi-Service Pilot Training Flight Training	CNATRAINST 1542.175D	1
b. Flight Training Instructions	CNATRA P-Pubs	various

2. Major Training Device

a. FTD/UTD quantity controlled by Naval Air Warfare Center Training Systems Division (NAWCTSD), Training Material Management Division, Inventory Control Branch (Code 5204).

b. Partial Task Trainer/Desktop Training Station quantity controlled by Naval Air Warfare Center Training Systems Division (NAWCTSD), Training Material Management Division, Inventory Control Branch (Code 5204).

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